Anchor: NGC4258 (Host of SN 1981K, SN 2014bc)

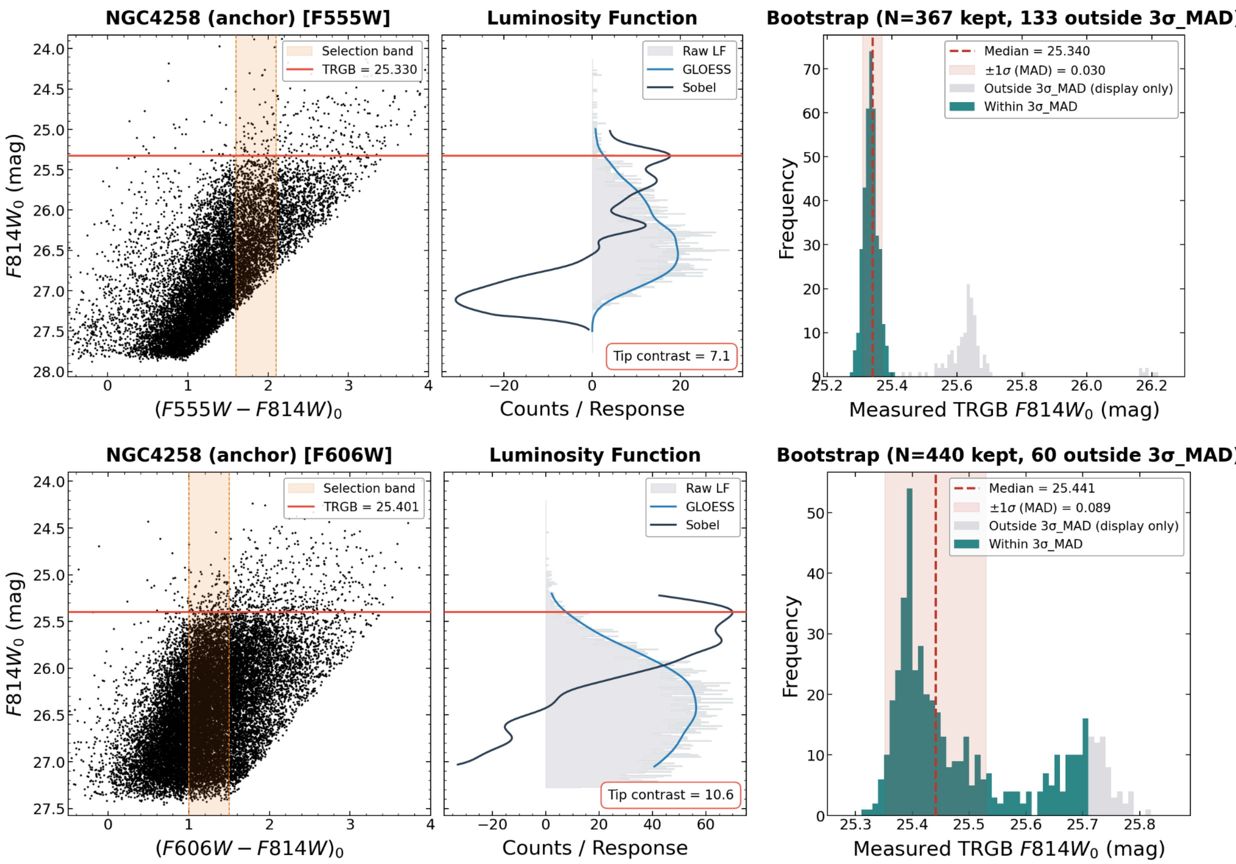
Date Processed: 2026-07-16 **Photometry Source:** EDD
Geometric Distance Modulus (μ_{maser}): 29.397 ± 0.032 mag

1. Initial Assessment

- **Data Quality:** Good. Clear RGB with some noise and a little AGB contamination. A bit messier in the F606W filter
- **Extinction:** $E(B-V) = [0.014]$

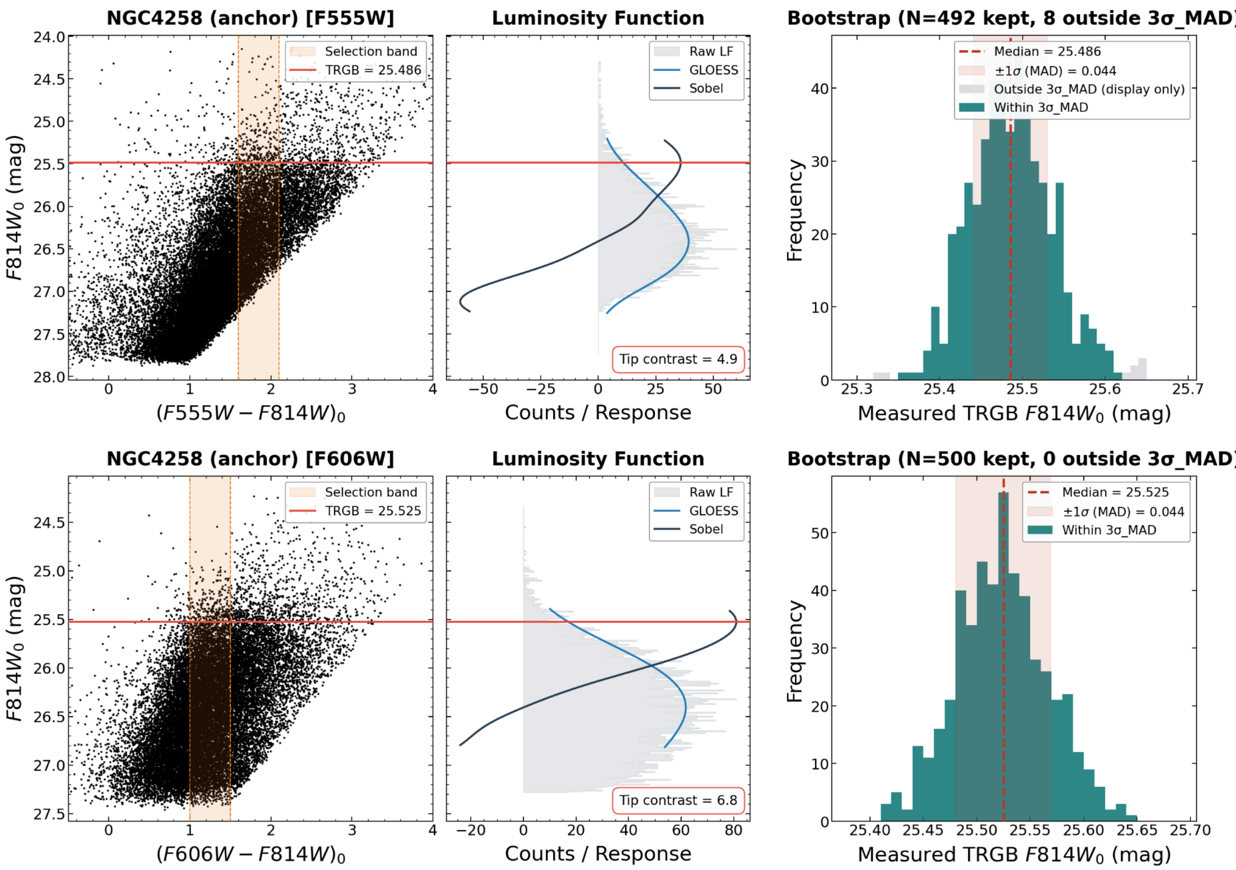
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
    - Increased to 0.18 to smooth out the Sobel filter and get rid of bimodality of Bootstrap graphs in both F555W and F606W filters
- **Color Selection (`color_lo`, `color_hi`):**
    - *F555W:* `[1.6]`, `[2.1]`: No need to change
    - *F606W:* `[1.0]`, `[1.5]`: No need to change
- **Spatial Clipping (`scale_factor`):** `[0.8]`
    - Decreased to 0.8 to get rid of the second peak outside 3σ_MAD in both F555W and F606W filters



3. Final Diagnostics

- **Tip Contrast:**
  - F555W: [4.9] *(Target > 3)*
  - F606W: [6.8] *(Target > 3)*
- **Bootstrap σ_{MAD}:**
  - F555W: [0.044] mag *(Target < 0.10)*
  - F606W: [0.044] mag *(Target < 0.10)*
- **Bootstrap Shape:**
  - F555W: Unimodal and narrow
  - F606W: Unimodal and narrow



4. Calibration Result

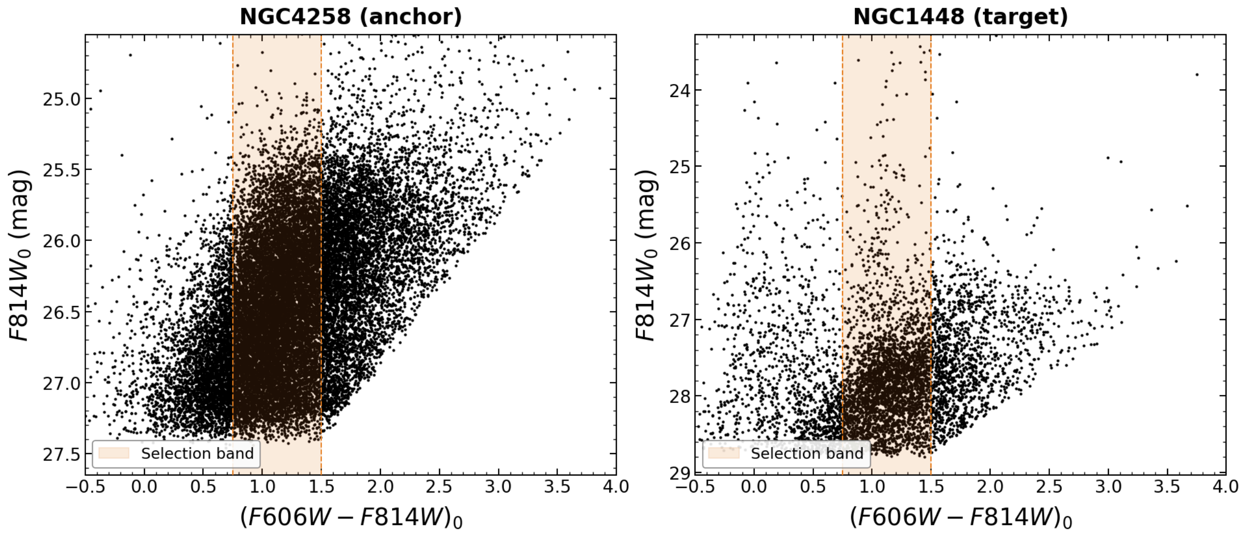
- **m_{TRGB} (Apparent):** `[25.505] ± [0.031]` mag
    - F5552:`[25.486] ± [0.044]` mag
    - F606W: `[25.525] ± [0.044]` mag
- **M_{TRGB} (Absolute Zero-Point):** `[-3.892] ± [0.046]` mag

Target #1: NGC1448 (Host of SN 2001e1, SN 2021pit)

Date Processed: 2026-07-16 **Photometry Source:** EDD

1. Initial Assessment

- **Data Quality:** Good. Somewhat sparse halo population; RGB is faintly defined. Light AGB contamination above the tip. High tip contrast and 2,284 RGB stars selected with N_faint of 1482 stars, so large sample size and low statistical noise
- **Extinction:** $E(B-V) = [0.013]$

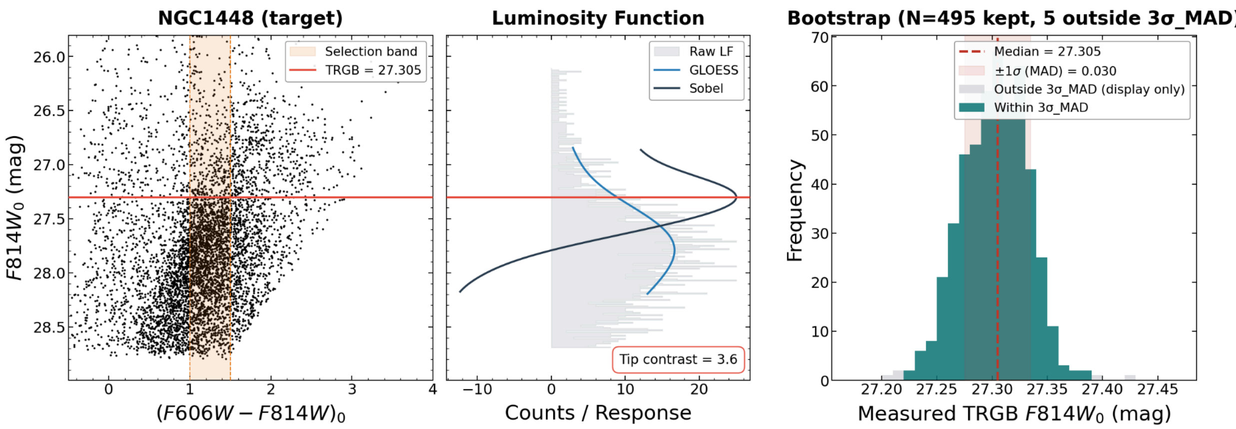


2. Parameter Choices & Justification

- **Smoothing (τ):** [0.18]
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** [1], [1.5]
  - No need to change from inherited anchor values in F606W filter
- **Spatial Clipping (`scale_factor`):** [0.8]
  - No need to change from inherited anchor values

3. Final Diagnostics

- **Tip Contrast:** [3.56] *(Target > 3)*
- **Bootstrap σ_{MAD}:** [0.030] mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** [1482] stars *(Target $\geq$ 400)*
- **Stability:** Very slightly unstable with a shift of 0.060 mag over τ range from 0.05 to 0.20 and a shift of 0.030 mag over color variations of +- 0.20 mag. Tried to decrease shift as much as possible by changing the weighting scheme to Poisson instead of Hatt and adding `MAG_BRIGHT_TARGET` of 26 *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

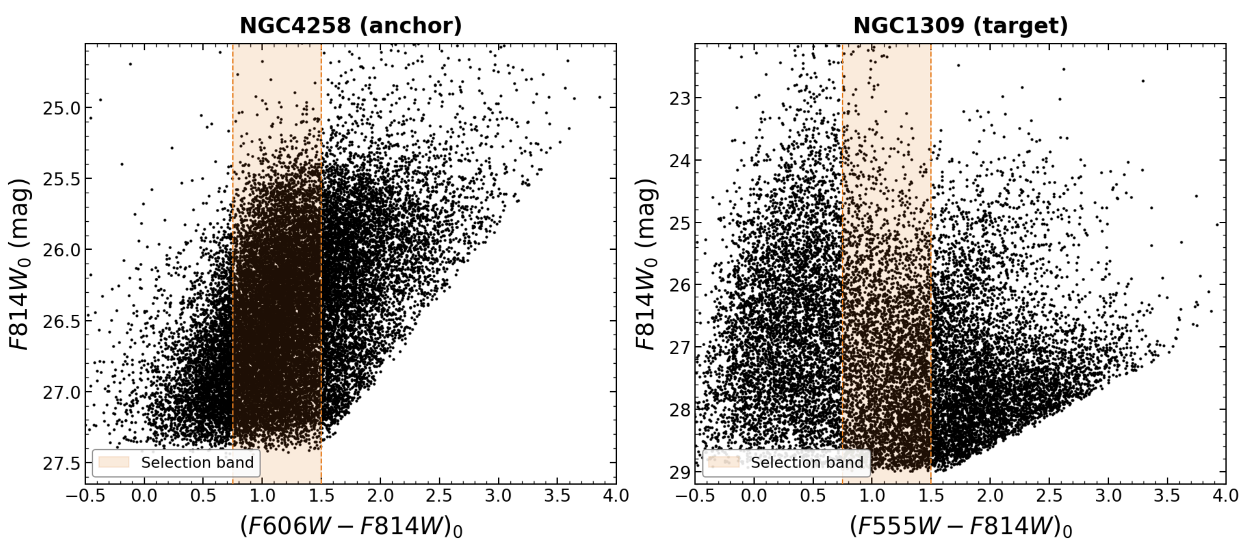
- **m_{TRGB} (Apparent):** [27.305] ± [0.030] mag
- **Final Distance (d):** [17.35] ± [0.44] Mpc

Target #2: NGC1309 (Host of SN 2002fk)

Date Processed: 2026-07-19 **Photometry Source:** EDD

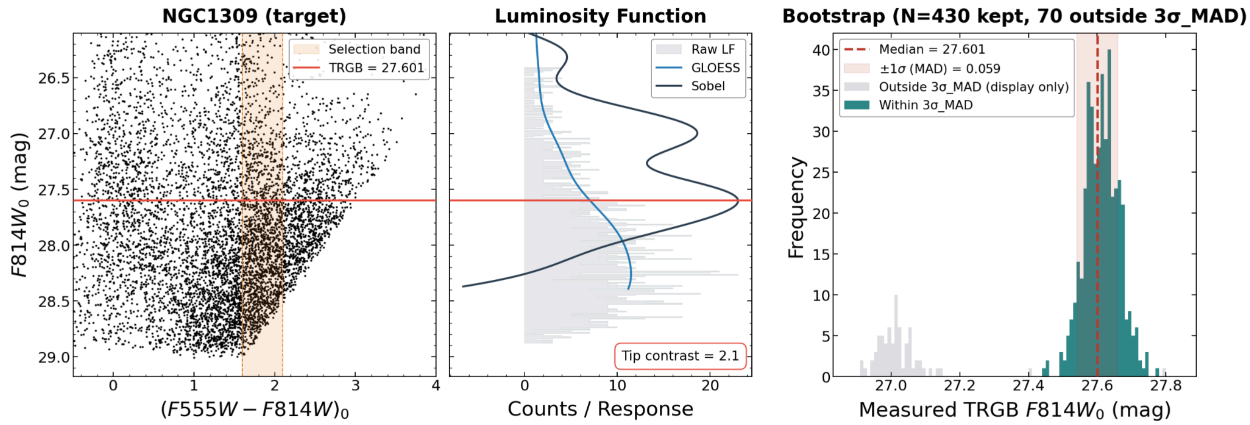
1. Initial Assessment

- **Data Quality:** Okay. RGB is faintly defined because of heavy AGB contamination but sufficient tip contrast of 3.06 and sample size with 897 selected RGB stars and N_{faint} of 625 stars
- **Extinction:** $E(B-V) = [0.035]$



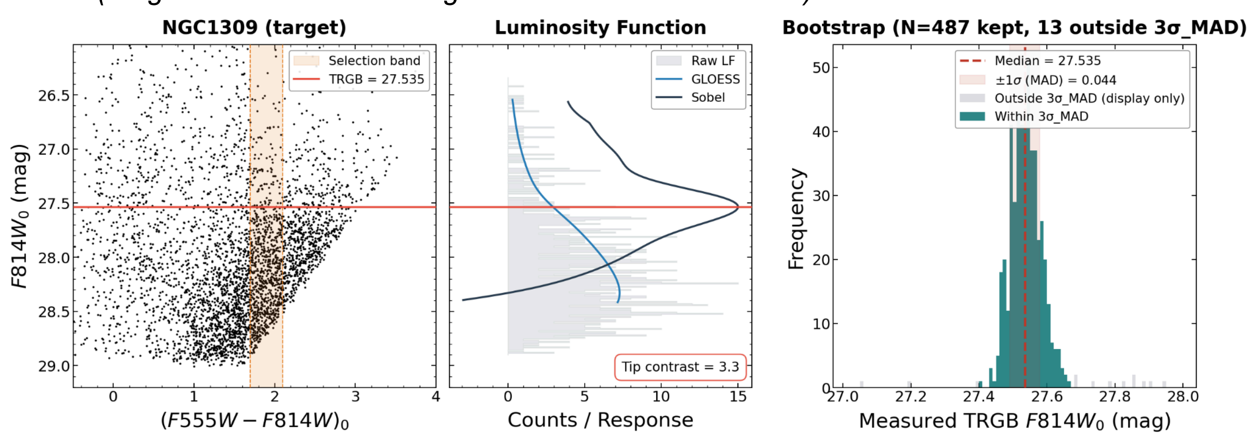
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
    - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.7]`, `[2.1]`
    - Tightened to lower instability as much as possible
- **Spatial Clipping (`scale_factor`):** `[1]`
    - Increased to get rid of bimodality in both Sobel filter and bootstrap graphs



3. Final Diagnostics

- **Tip Contrast:** `[3.28]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.044]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and very tight
- **N_{faint}:** `[625]` stars *(Target $\geq$ 400)*
- **Stability:** Very slightly unstable with a shift of 0.060 over τ range from 0.05 to 0.20 and with a shift of 0.040 mag over color variations of +- 0.20 mag. Decreased shifts as much as possible by tightening color window and adding `MAG_BRIGHT_TARGET` of 26 *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

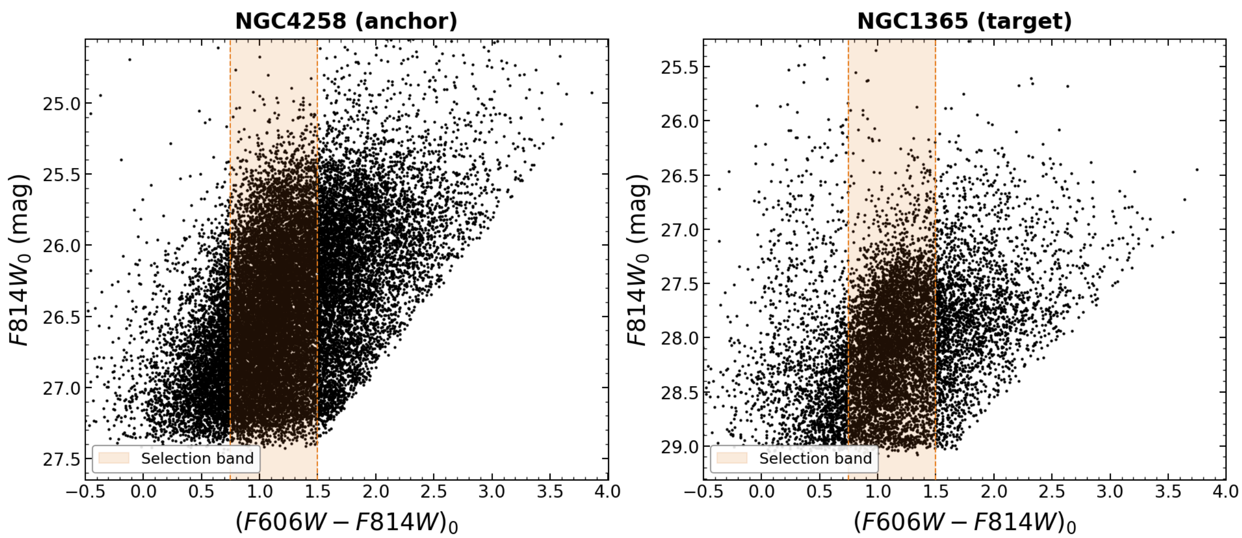
- **m_{TRGB} (Apparent):** `[27.535] ± [0.044]` mag
- **Final Distance (d):** `[19.29] ± [0.58]` Mpc

Target #3: NGC1365 (Host of SN 2012fr)

Date Processed: 2026-07-19 **Photometry Source:** EDD

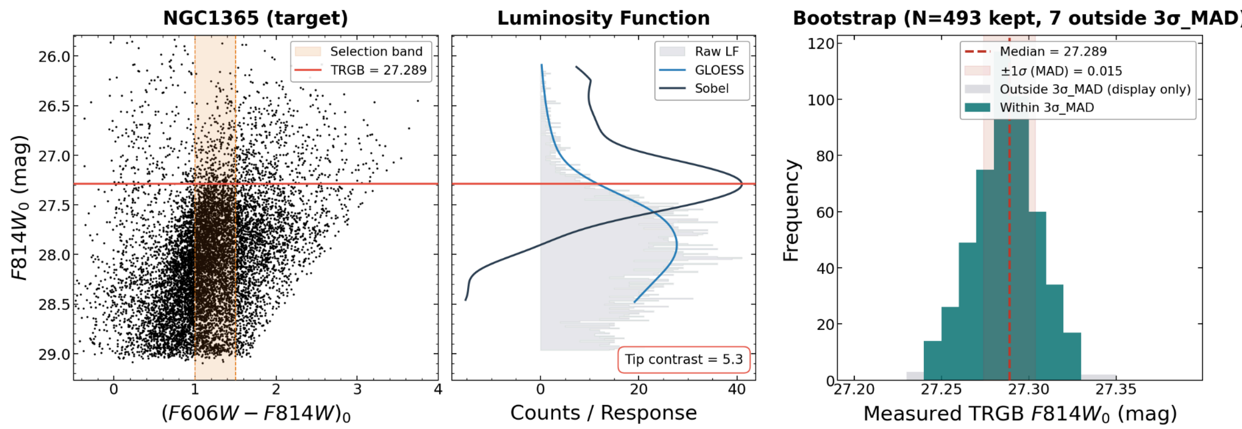
1. Initial Assessment

- **Data Quality:** Good. Somewhat clear RGB with a little AGB contamination. Well-populated with 3,957 stars selected. Very high tip contrast of 5.31 and very good star counts below the edge with N_{faint} of 2465.
- **Extinction:** $E(B-V) = [0.018]$4



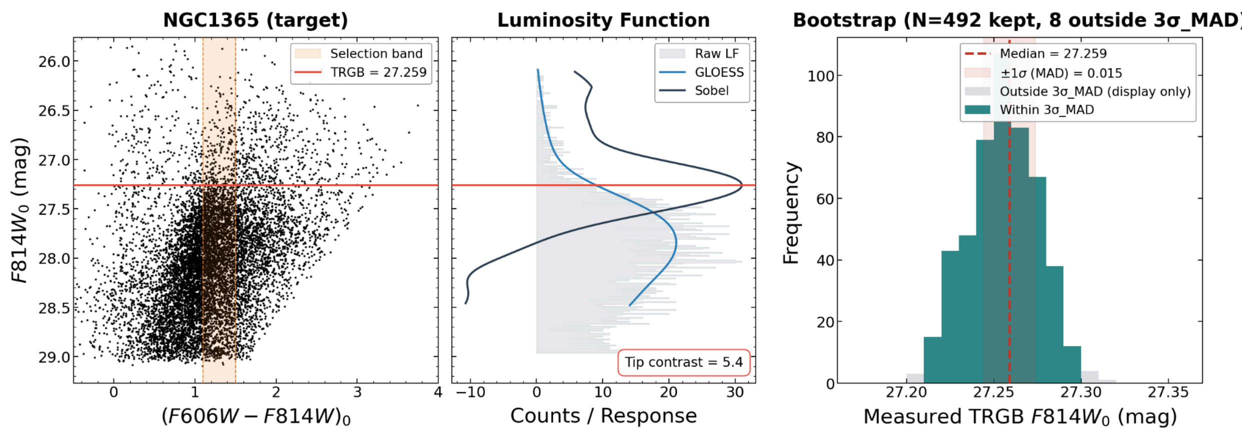
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.1]`, `[1.5]`
  - Tightened to lower instability
- **Spatial Clipping (`scale_factor`):** `[0.8]`
  - No need to change from inherited anchor values



3. Final Diagnostics

- **Tip Contrast:** `[5.42]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.015]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and very tight
- **N_{faint}:** `[1874]` stars *(Target $\geq$ 400)*
- **Stability**: Slightly unstable with a shift of 0.080 over τ range from 0.05 to 0.20 and with a shift of 0.070 mag over color variations of `$\pm$` 0.20 mag. Tried to decrease this by increasing **`color_lo`**. (Target < 0.03 over `$\tau$` range and over color variations)



4. Distance Result

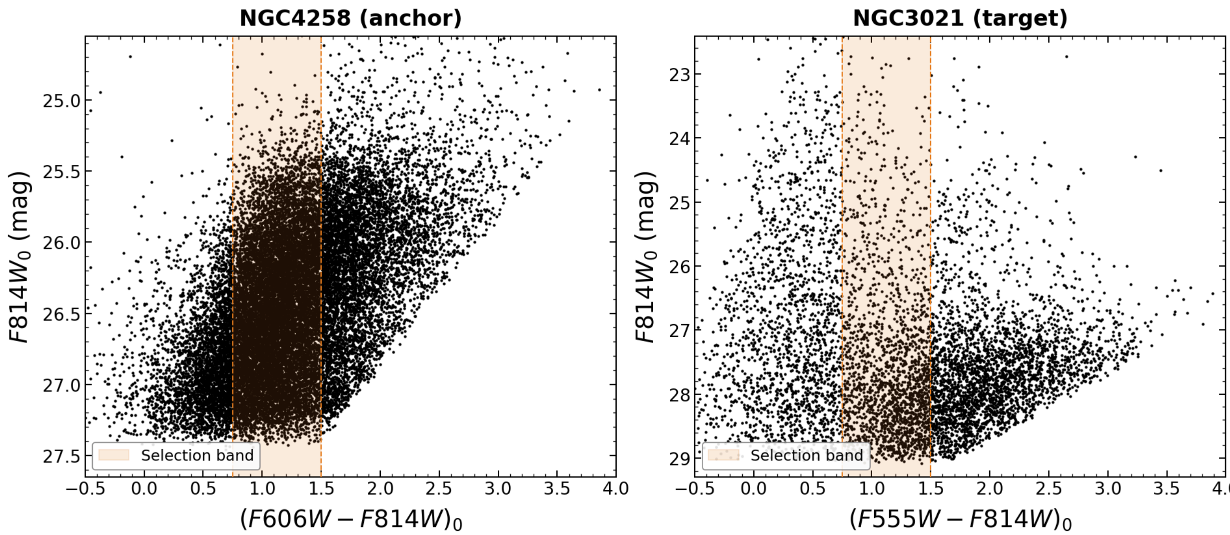
- **m_{TRGB} (Apparent):** `[27.259] ± [0.015]` mag
- **Final Distance (d):** `[16.99] ± [0.39]` Mpc

Target #4: NGC3021 (Host of SN 1995al)

Date Processed: 2026-07-19 **Photometry Source:** EDD

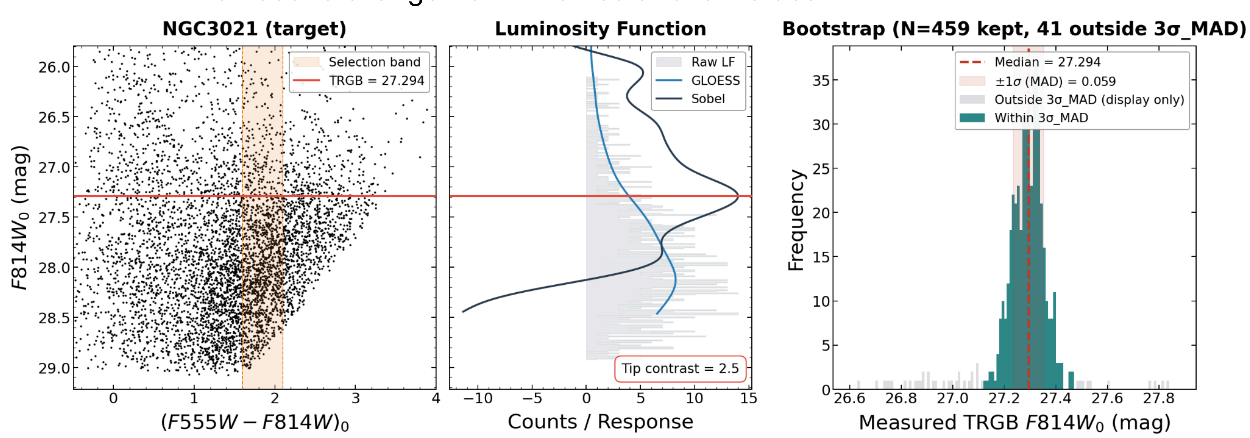
1. Initial Assessment

- **Data Quality:** Poor. RGB is faintly defined because of heavy AGB contamination Insufficient tip contrast of 2.57 but good sample size with 1,313 RGB stars selected and N_{faint} of 706 stars.
- **Extinction:** $E(B-V) = [0.014]$



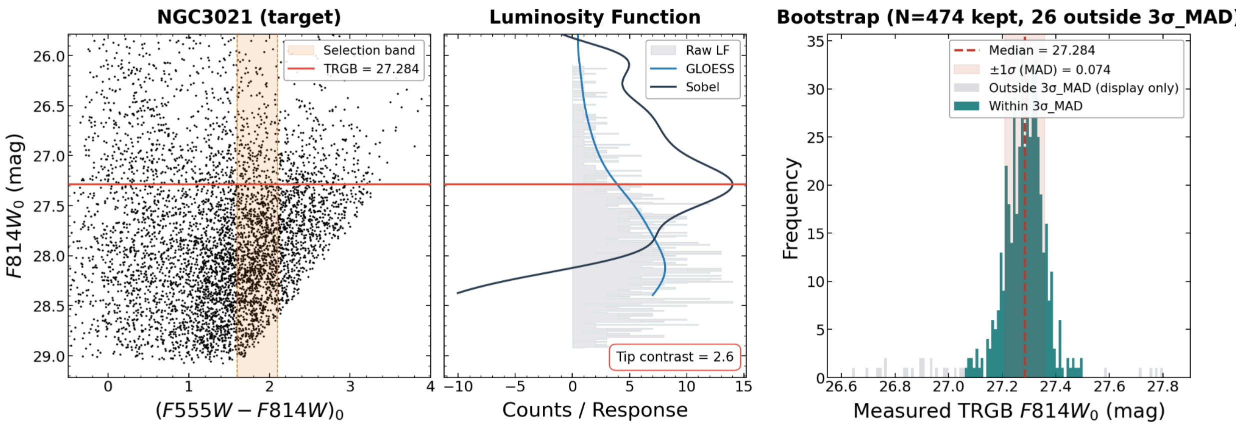
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.20]`
    - Increased to make sure N_{faint} > 400
- **Color Selection (`color_lo`, `color_hi`):** `[1.6]`, `[2.1]`
    - No need to change from inherited anchor values in F555W filter
- **Spatial Clipping (`scale_factor`):** `[0.8]`
    - No need to change from inherited anchor values



3. Final Diagnostics

- **Tip Contrast:** `[2.57]` *(Target > 3)*
    - Tried to increase by tightening **`color_lo`** and **`color_hi`** and changing **`scale_factor`** but could not do so without also lowering N_{faint} below 400
- **Bootstrap σ_{MAD}:** `[0.074]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and very tight
- **N_{faint}:** `[706]` stars *(Target $\geq$ 400)*
- **Stability:** Slightly unstable with a shift of 0.020 over τ range from 0.05 to 0.20 and shift of 0.090 over color variations of $\pm$ 0.20 mag. Tried to decrease this by changing the weighing scheme from Hatt to Poisson but could not get any lower. *(Target < 0.03 over τ range and over color variations)*



4. Distance Result

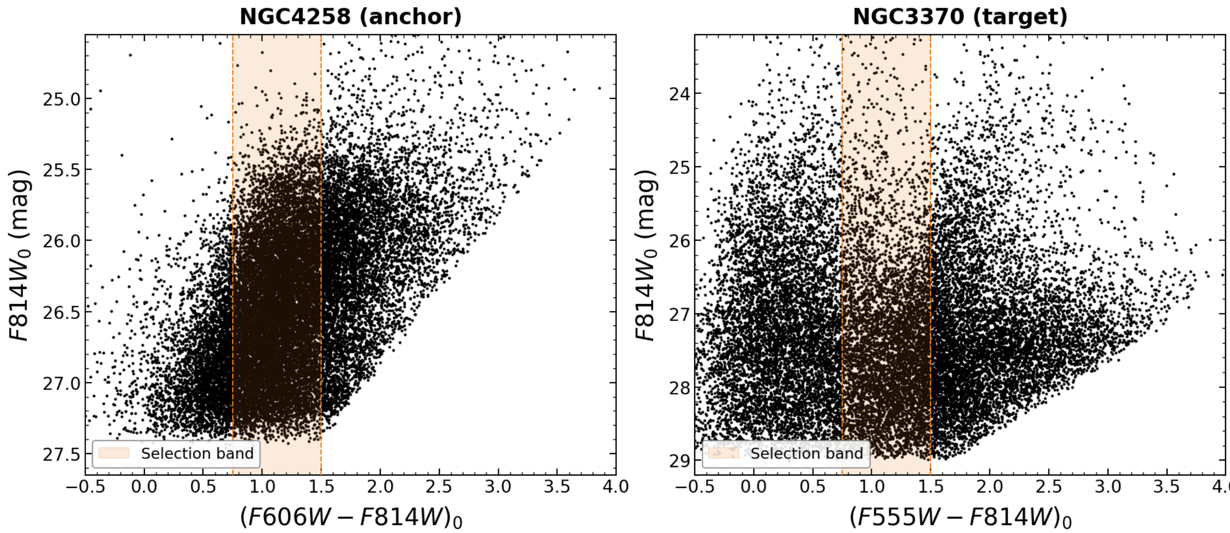
- **m_{TRGB} (Apparent):** `[27.284] ± [0.074]` mag
- **Final Distance (d):** `[17.18] ± [0.69]` Mpc

Target #5: NGC3370 (Host of SN 1994ae)

Date Processed: 2026-07-19 **Photometry Source:** EDD

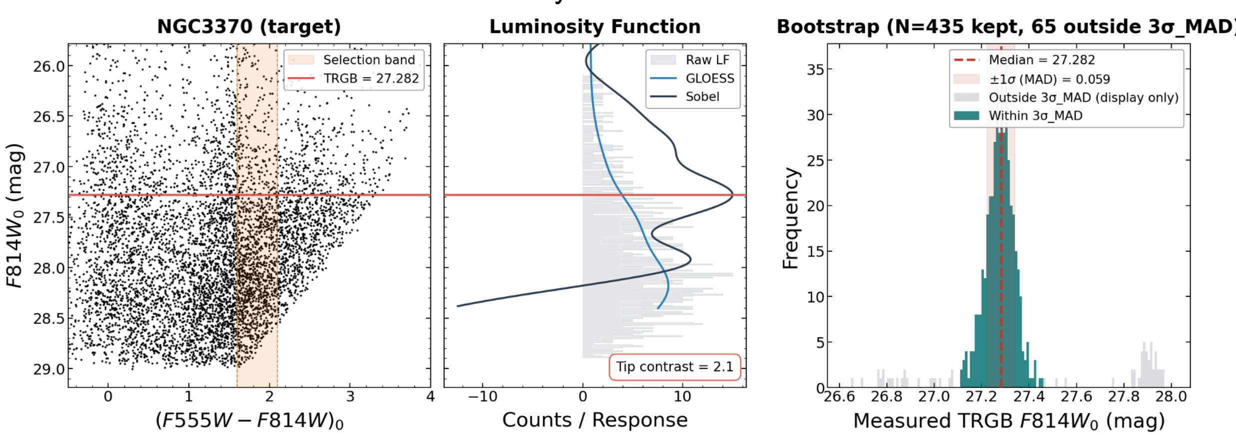
1. Initial Assessment

- **Data Quality:** Poor. RGB is barely defined because of heavy AGB contamination and lots of bluer, younger stars. Insufficient tip contrast but good sample size with N_{faint of 847 stars.
- **Extinction:** $E(B-V) = [0.031]$



2. Parameter Choices & Justification

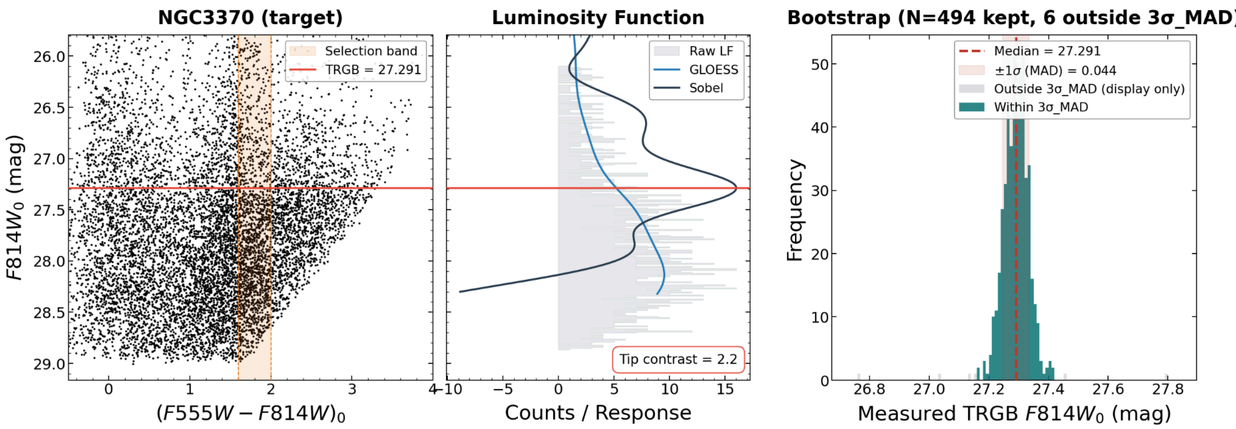
- **Smoothing (τ):** `[0.20]`
  - Increased to make sure N_{faint} > 400
- **Color Selection (`color_lo`, `color_hi`):** `[1.6]`, `[2.0]`
  - Tightened to increase tip contrast as much as possible and lower instability
- **Spatial Clipping (`scale_factor`):** `[0.7]`
  - Decreased to lower instability



3. Final Diagnostics

- **Tip Contrast:** `[2.16]` *(Target > 3)*
  - Tried to increase by tightening `color_lo` and `color_hi` and changing `scale_factor` but could not do so without also lowering N_{faint} below 400 or making highly unstable
- **Bootstrap σ_{MAD}:** `[0.044]` mag *(Target < 0.10)*

- **Bootstrap Shape:** Unimodal and very tight
- **N_{faint}:** `[847]` stars *(Target $\geq$ 400)*
- **Stability:** Slightly unstable with a shift of 0.060 over τ range from 0.05 to 0.20 and a shift of 0.030 over color variations of $\pm$ 0.20 mag. Decreased this as much as possible by decreasing **`scale_factor`** and tightening color limits *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

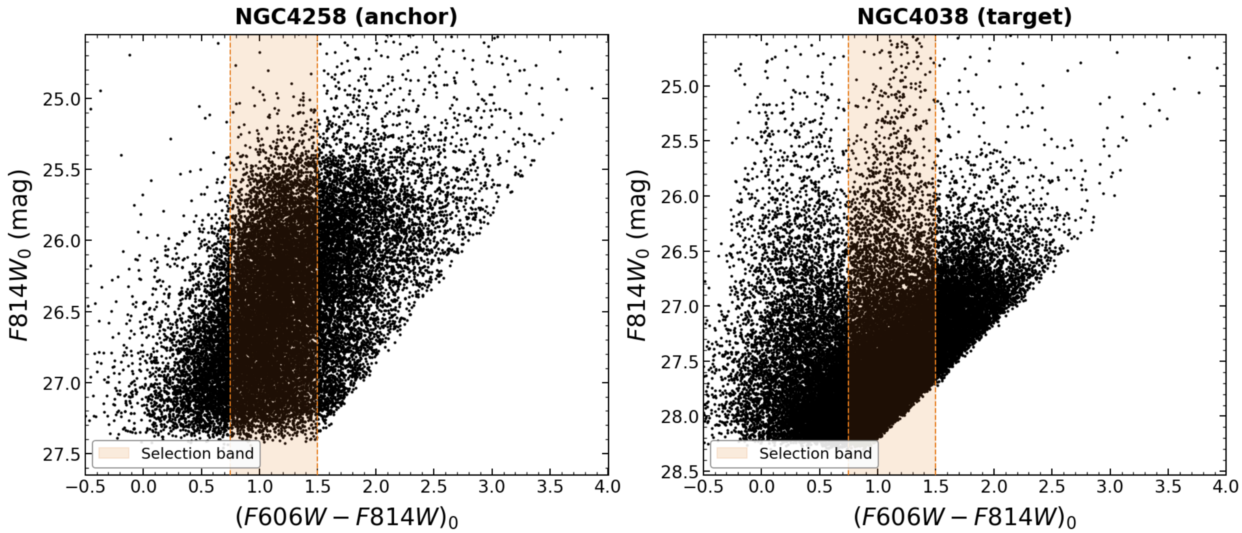
- **m_{TRGB} (Apparent):** `[27.291] ± [0.044]` mag
- **Final Distance (d):** `[17.24] ± [0.52]` Mpc

Target #6: NGC4038 (Host of SN 2007sr)

Date Processed: 2026-07-20 **Photometry Source:** EDD

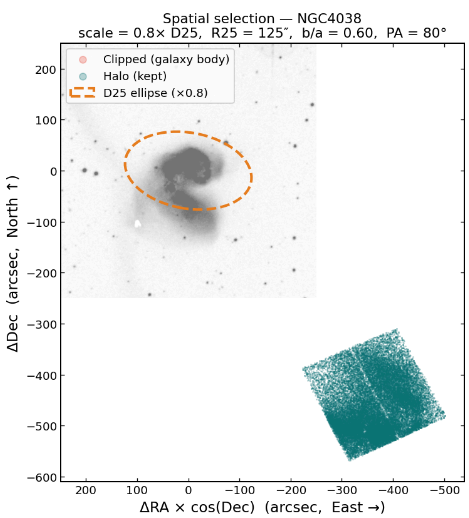
1. Initial Assessment

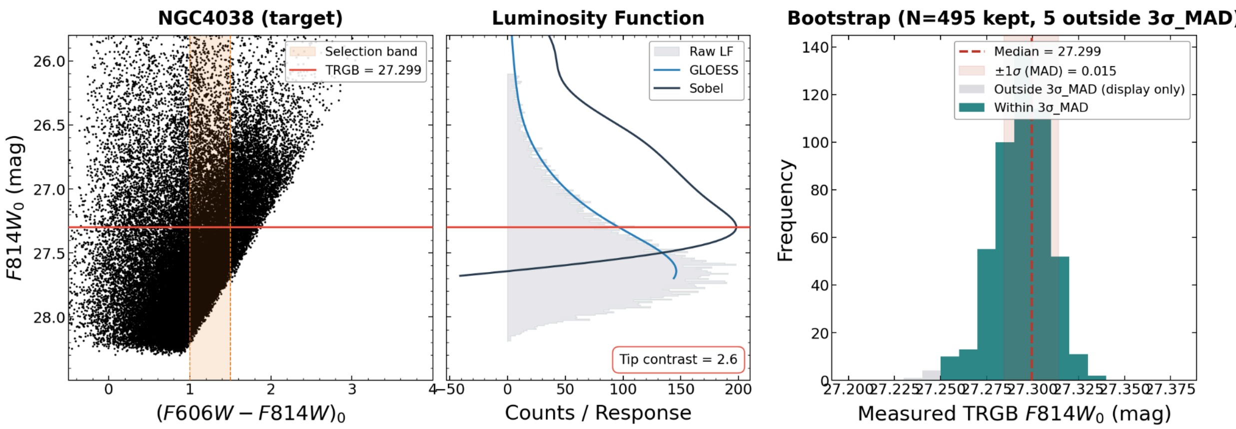
- **Data Quality:** Poor. Very well defined RGB with little AGB contamination. Massive sample size with 14,148 RGB stars selected and N_{faint} of 10,402 stars but insufficient tip contrast of 2.80.
- **Extinction:** $E(B-V) = [0.042]$



2. Parameter Choices & Justification

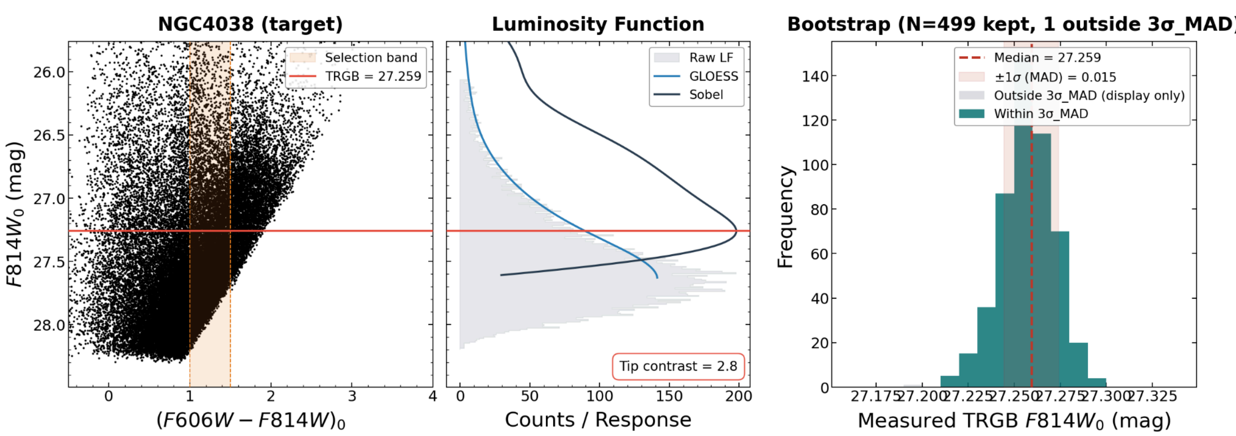
- **Smoothing (τ):** `[0.20]`
  - Increased from 0.18 to try and increase tip contrast
- **Color Selection (`color_lo`, `color_hi`):** `[1.0]`, `[1.5]`
  - No need to change from inherited anchor values, tried to tighten but decreased tip contrast
- **Spatial Clipping (`scale_factor`):** `[0.8]`
  - No need to change from inherited anchor values, as seen in spatial selection graph below; `scale_factor` needs to be > 4 to have any impact, and at > 4, there was no significant impact





3. Final Diagnostics

- **Tip Contrast:** [2.80] *(Target > 3)*
  - Tried to increase by tightening **color_lo** and **color_hi** and changing **scale_factor** but could not do so. Was able to achieve tip contrast > 3 by crossing the limit of 1.5 mag for color_hi but result was still highly unstable
- **Bootstrap σ_{MAD}:** [0.015] mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and very tight
- **N_{faint}:** [10402] stars *(Target $\geq$ 400)*
- **Stability:** Highly unstable with a shift of 0.150 over τ range from 0.05 to 0.20 and a shift of 0.290 over color variations of $\pm$ 0.20 mag. Decreased this as much as possible by decreasing **scale_factor** and widening color limits but that did not have an impact *(Target < 0.03 over* τ *range and over color variations)*



4. Distance Result

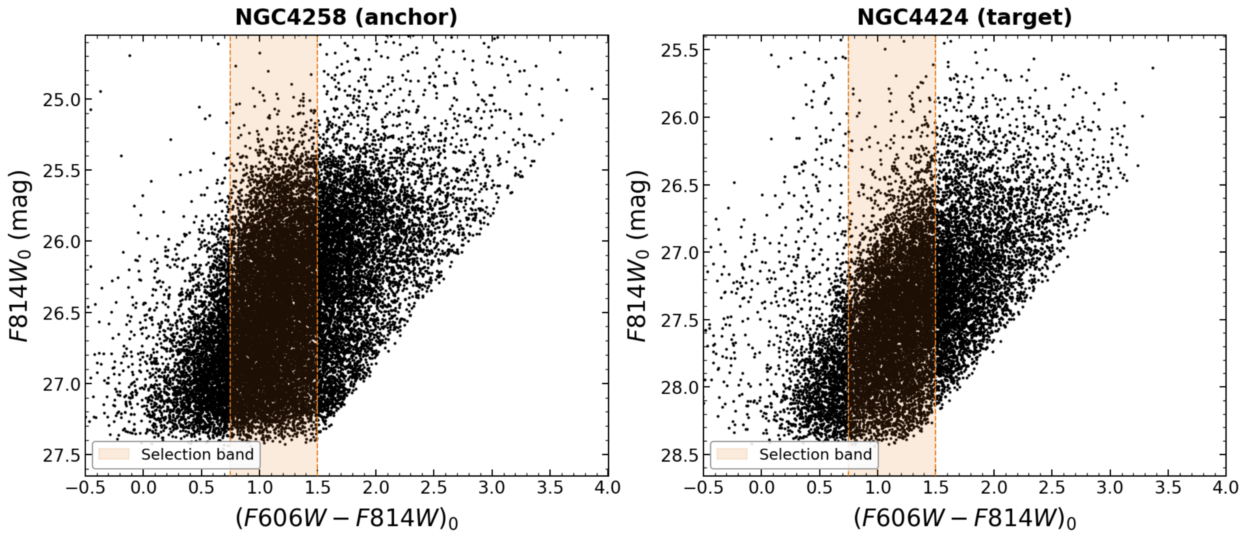
- **m_{TRGB} (Apparent):** [27.259] ± [0.015] mag
- **Final Distance (d):** [16.99] ± [0.40] Mpc

Target #7: NGC4424 (Host of SN 2012cg)

Date Processed: 2026-07-20 **Photometry Source:** EDD

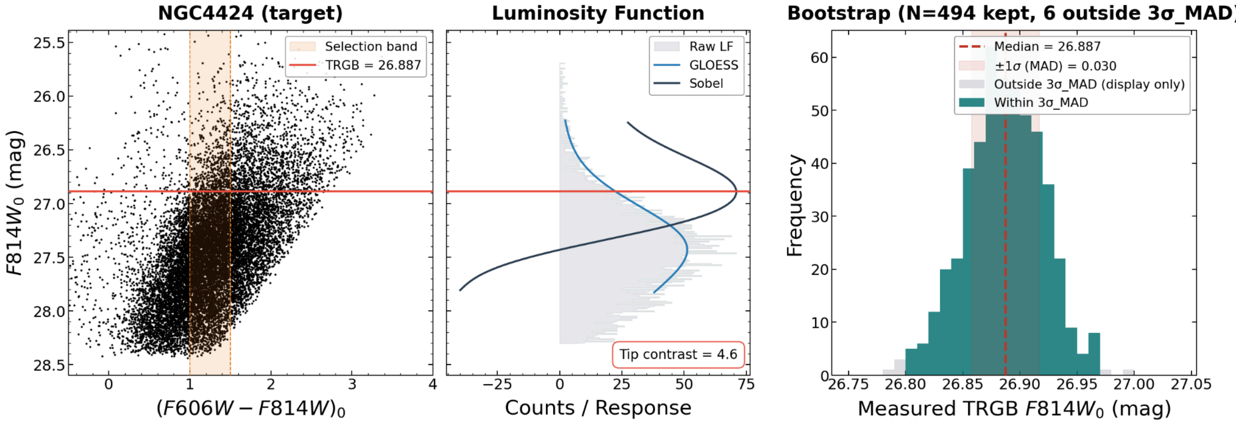
1. Initial Assessment

- **Data Quality:** Excellent. Well defined RGB with very little AGB contamination. Very high tip contrast and good sample size with 3,918 RGB selected stars and N_{faint} of 2,870 stars.
- **Extinction:** $E(B-V) = [0.020]$



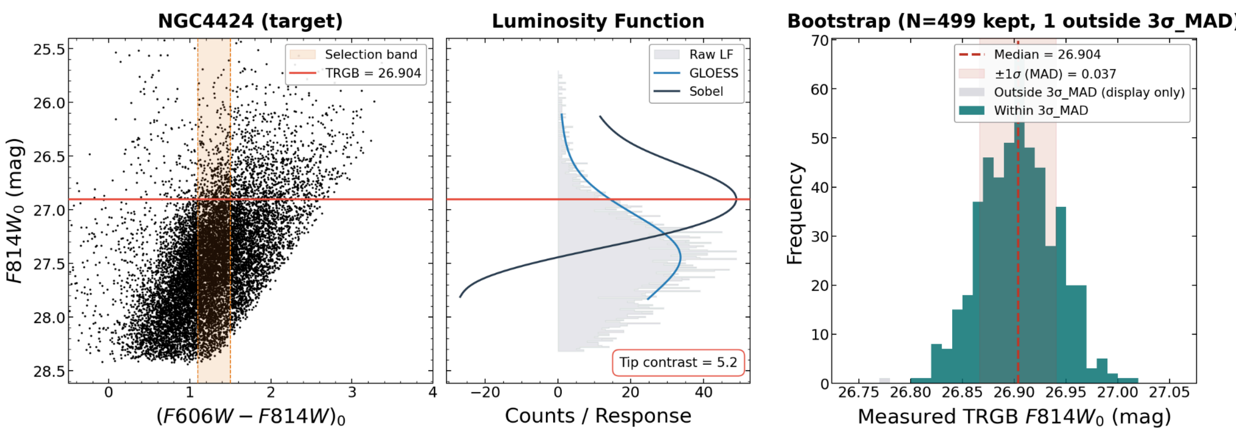
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.1]`, `[1.5]`
  - Tightened to lower instability
- **Spatial Clipping (`scale_factor`):** `[1]`
  - Increased to lower instability



3. Final Diagnostics

- **Tip Contrast:** `[5.16]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.037]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** `[2870]` stars *(Target $\geq$ 400)*
- **Stability:** Unstable with a shift of 0.050 over τ range from 0.05 to 0.20 and a shift of 0.130 over color variations of $\pm$ 0.20 mag. Decreased this as much as possible by increasing **`scale_factor`** and widening color limits but could not get either below 0.03 *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

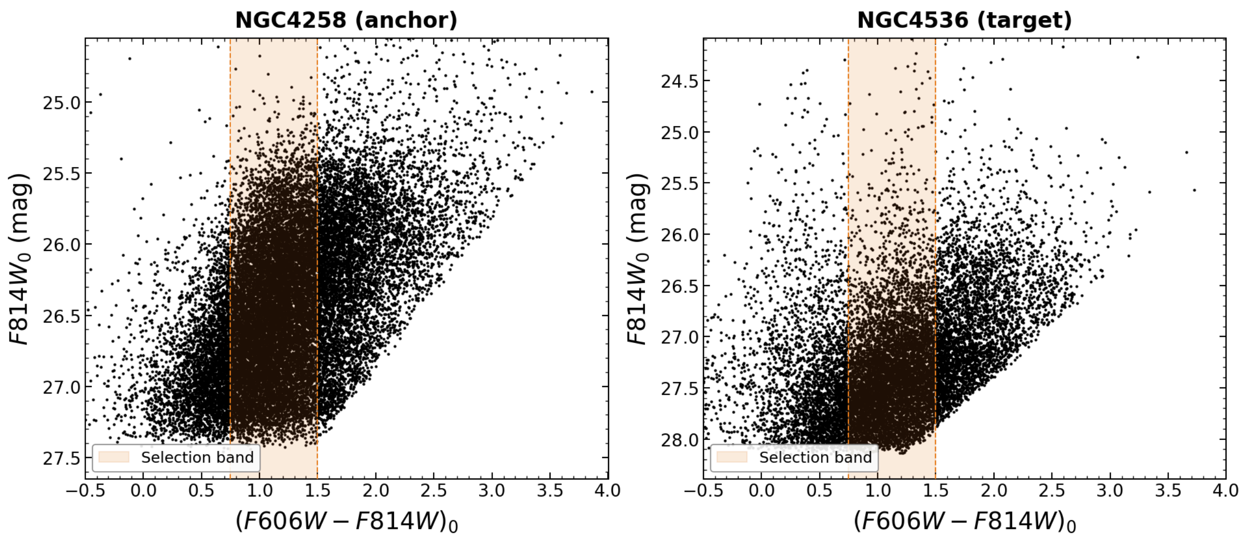
- **m_{TRGB} (Apparent):** `[26.904] ± [0.037]` mag
- **Final Distance (d):** `[14.42] ± [0.40]` Mpc

Target #8: NGC4536 (Host of SN 1981B)

Date Processed: 2026-07-20 **Photometry Source:** EDD

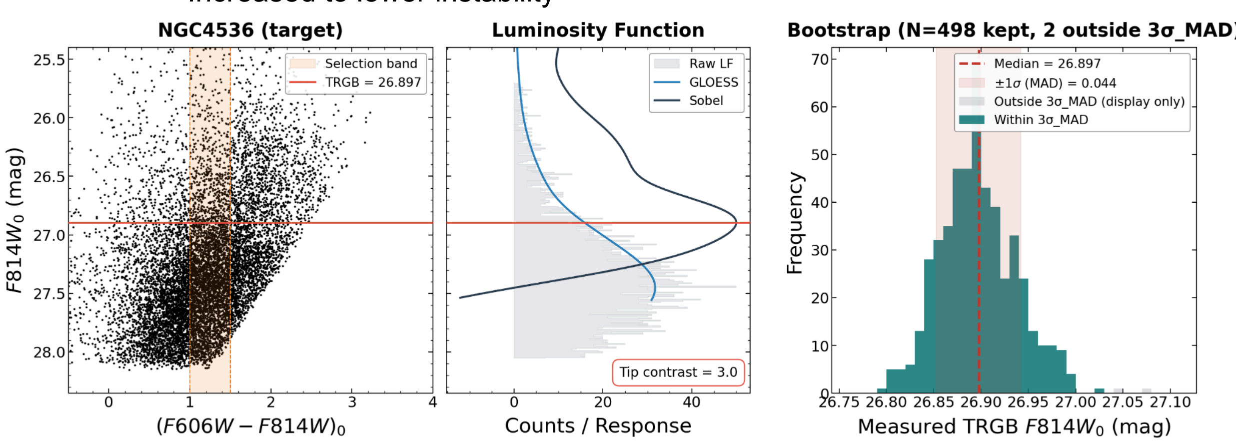
1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.018]$



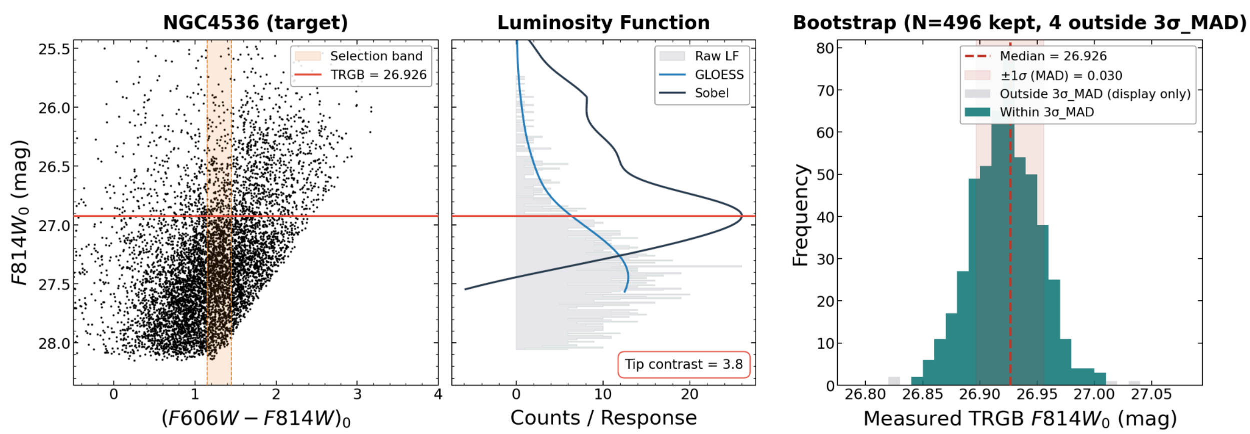
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
    - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.15]`, `[1.45]`
    - Tightened to lower instability
- **Spatial Clipping (`scale_factor`):** `[1]`
    - Increased to lower instability



3. Final Diagnostics

- **Tip Contrast:** `[3.75]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.030]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** `[1148]` stars *(Target $\geq$ 400)*
- **Stability:** Unstable with a shift of 0.120 over color variations of +- 0.20 mag. Decreased this as much as possible by increasing **`scale_factor`** and tightening color limits but could not get it below 0.03 *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

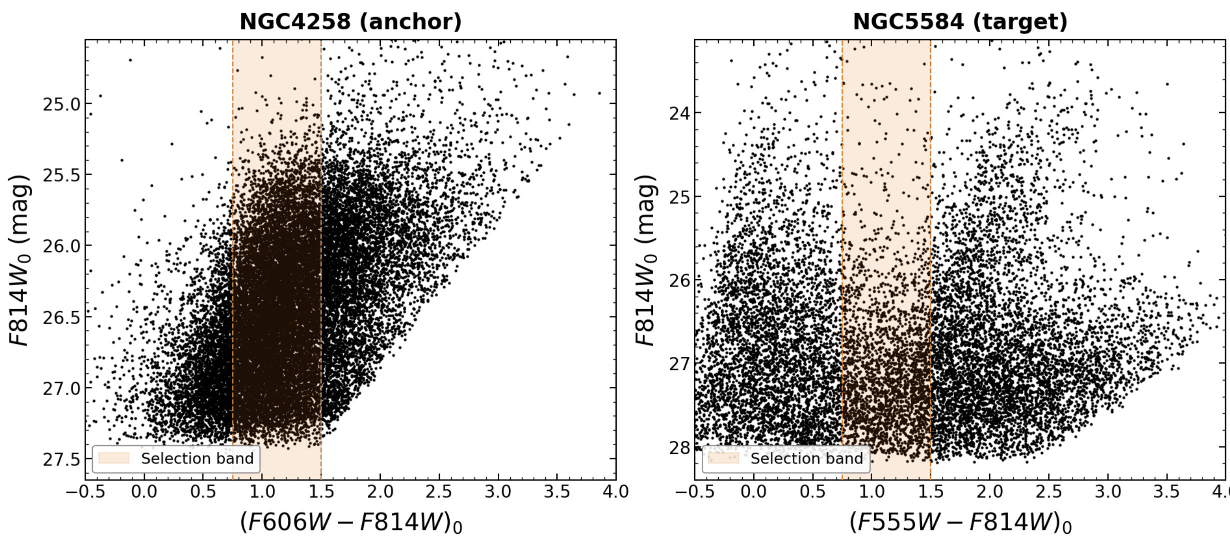
- **m_{TRGB} (Apparent):** `[26.926] ± [0.030]` mag
- **Final Distance (d):** `[14.58] ± [0.37]` Mpc

Target #9: NGC5584 (Host of SN 1981B)

Date Processed: 2026-07-20 **Photometry Source:** EDD

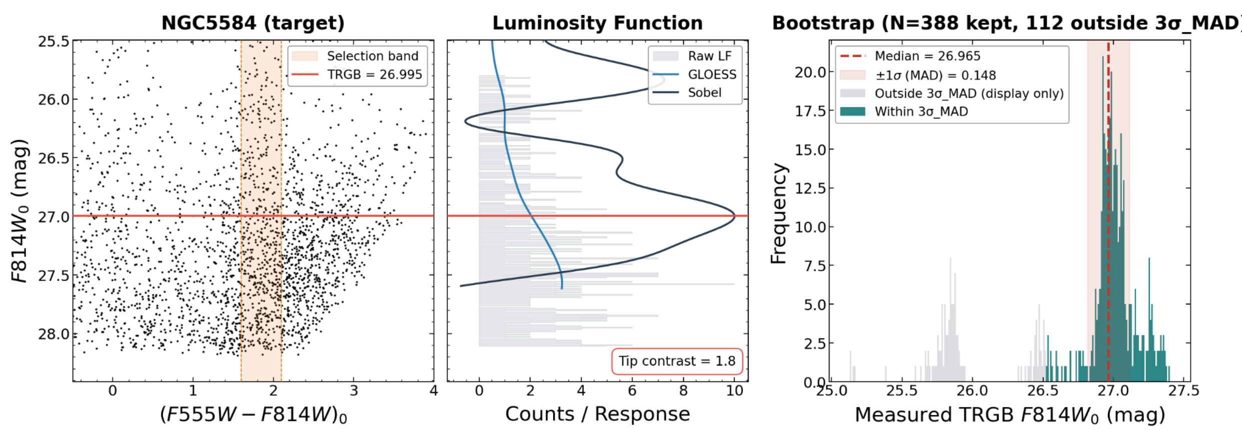
1. Initial Assessment

- **Data Quality:** Bad, high contamination
- **Extinction:** $E(B-V) = [0.035]$



2. Parameter Choices & Justification

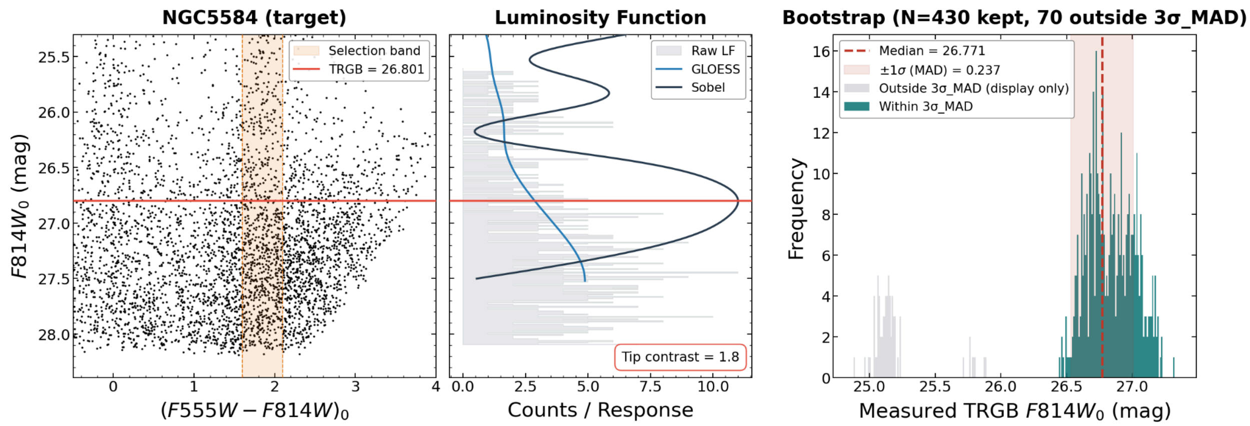
- **Smoothing (τ):** `[0.25]`
  - Increased to try and get tip contrast > 3 and get rid of bimodality for both Sobel filter and bootstrap graphs
- **Color Selection (`color_lo`, `color_hi`):** `[1.6]`, `[2.1]`
  - No need to change from inherited anchor values for F555W filter
- **Spatial Clipping (`scale_factor`):** `[0.7]`
  - Decreased to make sure N_faint > 400



3. Final Diagnostics

- **Tip Contrast:** `[1.81]` *(Target > 3)*
  - Tried to increase by tightening **`color_lo`** and **`color_hi`** and changing **`scale_factor`** but could not do so without getting N_faint < 400.
- **Bootstrap σ_{MAD}:** `[0.163]` mag *(Target < 0.10)*
  - Could not decrease to get below 0.10

- **Bootstrap Shape:** Somewhat bimodal and wide
- **N_{faint}:** `[429]` stars *(Target $\geq$ 400)*
- **Stability:** Very unstable with a shift of 1.600 over τ range from 0.05 to 0.20 and a shift of 0.120 over color variations of +- 0.20 mag. Decreased this as much as possible by increasing **`scale_factor`** and tightening color limits but could not get it below 0.03 for either *(Target < 0.03 over tau range and over color variations)*



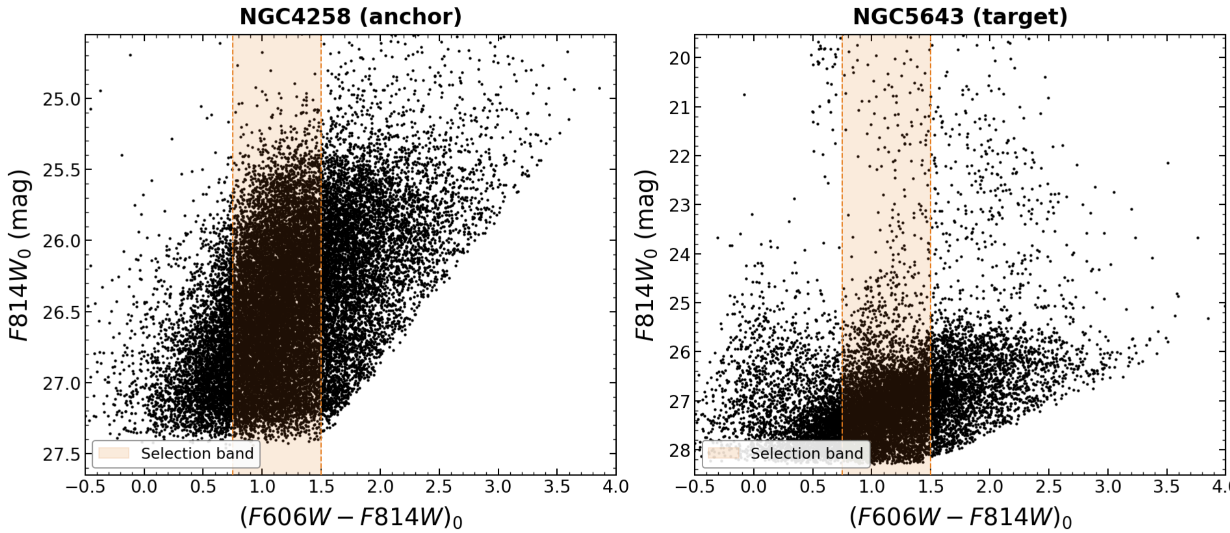
The TRGB for this galaxy is ultimately unresolvable, so I have excluded it from primary distance modulus synthesis due to very large instability. Flagged as a low-contrast non-detection.

Target #10: NGC5643 (Host of SN 2013aa, SN 2017cbv)

Date Processed: 2026-07-20 **Photometry Source:** EDD

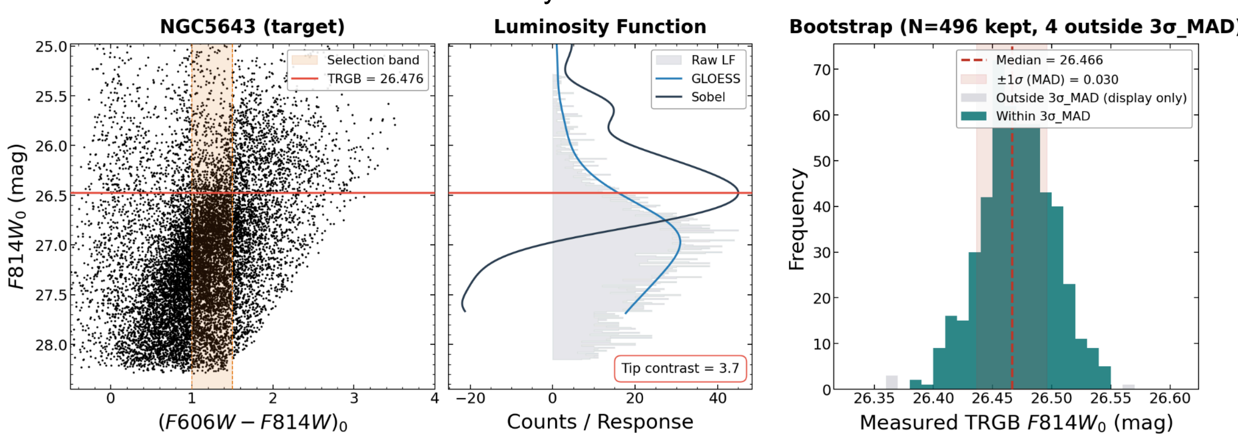
1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.166]$ Very high, abnormal



2. Parameter Choices & Justification

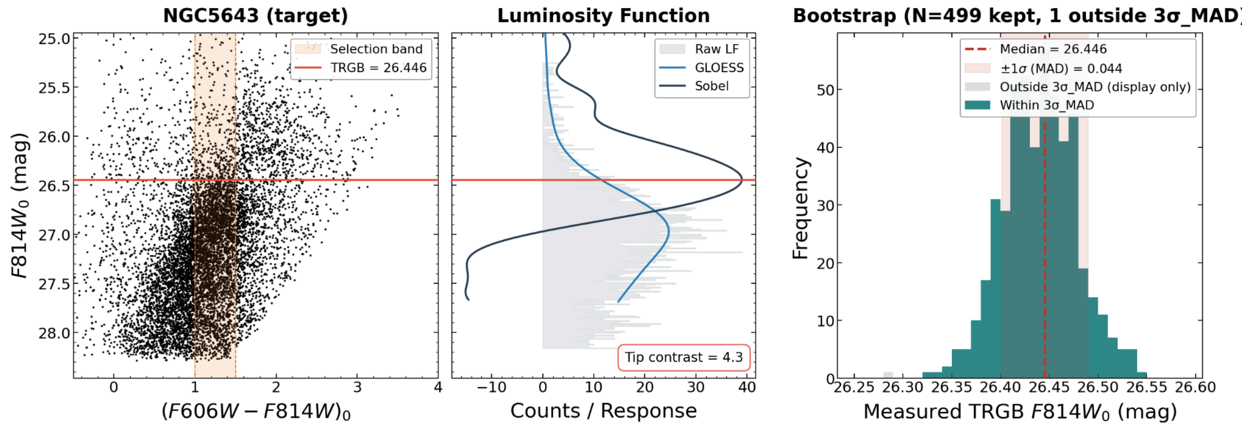
- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.0]`, `[1.5]`
  - No need to change from inherited anchor values
- **Spatial Clipping (`scale_factor`):** `[1]`
  - Increased to lower instability



3. Final Diagnostics

- **Tip Contrast:** `[4.31]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.044]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** `[2154]` stars *(Target $\geq$ 400)*
- **Stability:** Very slightly unstable with a shift of 0.050 over τ range from 0.05 to 0.20 and a shift of 0.060 over color variations of +- 0.20 mag. Decreased this as much as

possible by increasing **`scale_factor`** but could not get it below 0.03 *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

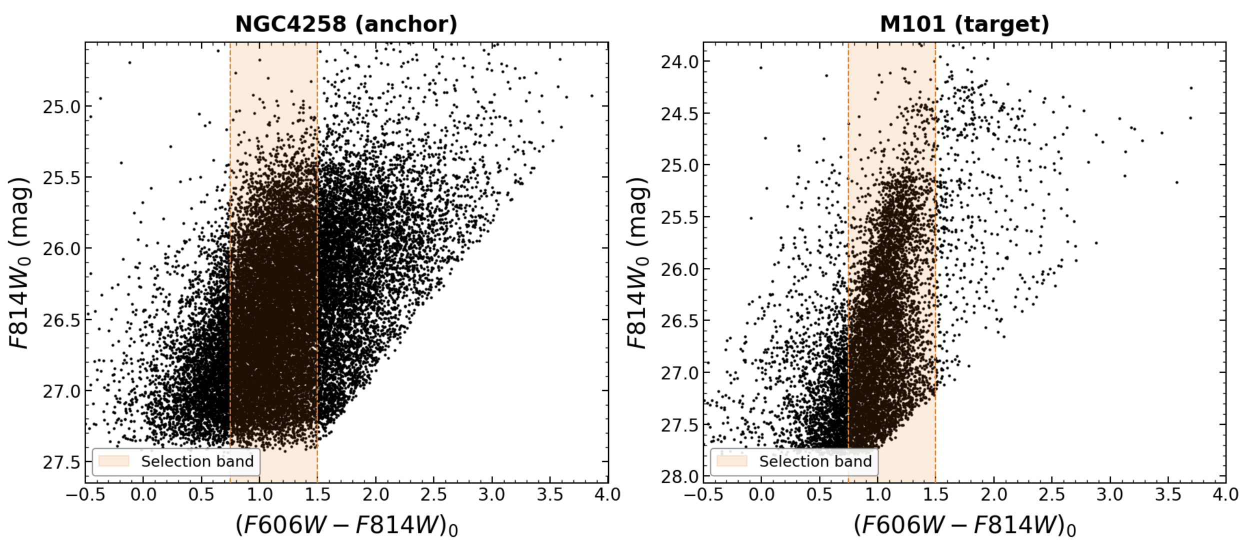
- **m_{TRGB} (Apparent):** `[26.446] ± [0.044]` mag
- **Final Distance (d):** `[11.68] ± [0.46]` Mpc

Target #11: M101 (Host of SN 2011fe)

Date Processed: 2026-07-21 **Photometry Source:** EDD

1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.007]$ Very low

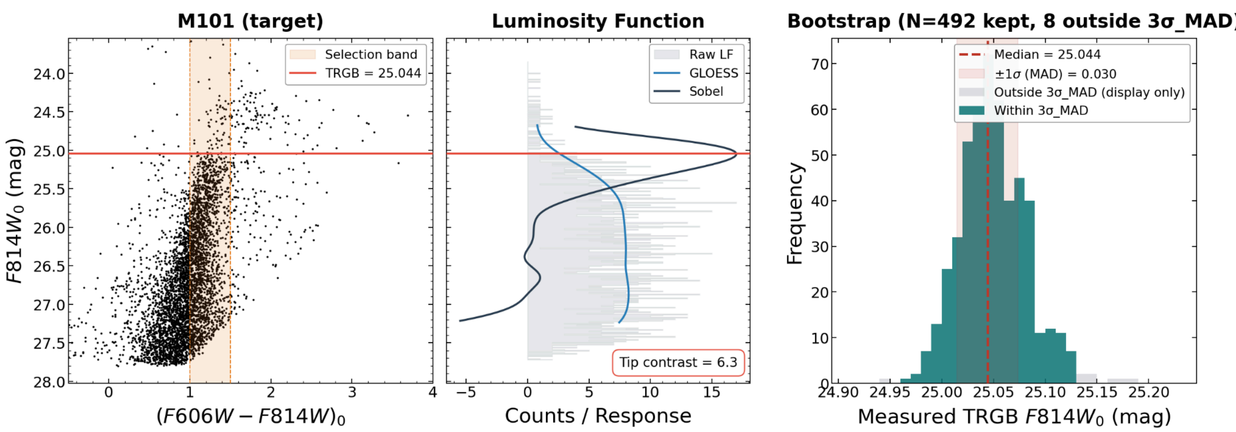


2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.0]`, `[1.5]`
  - No need to change from inherited anchor values
- **Spatial Clipping (`scale_factor`):** `[0.8]`
  - No need to change from inherited anchor values

3. Final Diagnostics

- **Tip Contrast:** `[6.26]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.030]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** `[660]` stars *(Target $\geq$ 400)*
- **Stability:** Stable with shifts over τ and over color variations of less than 0.03 mag *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

- **m_{TRGB} (Apparent):** `[25.044] ± [0.030]` mag
- **Final Distance (d):** `[6.13] ± [0.16]` Mpc

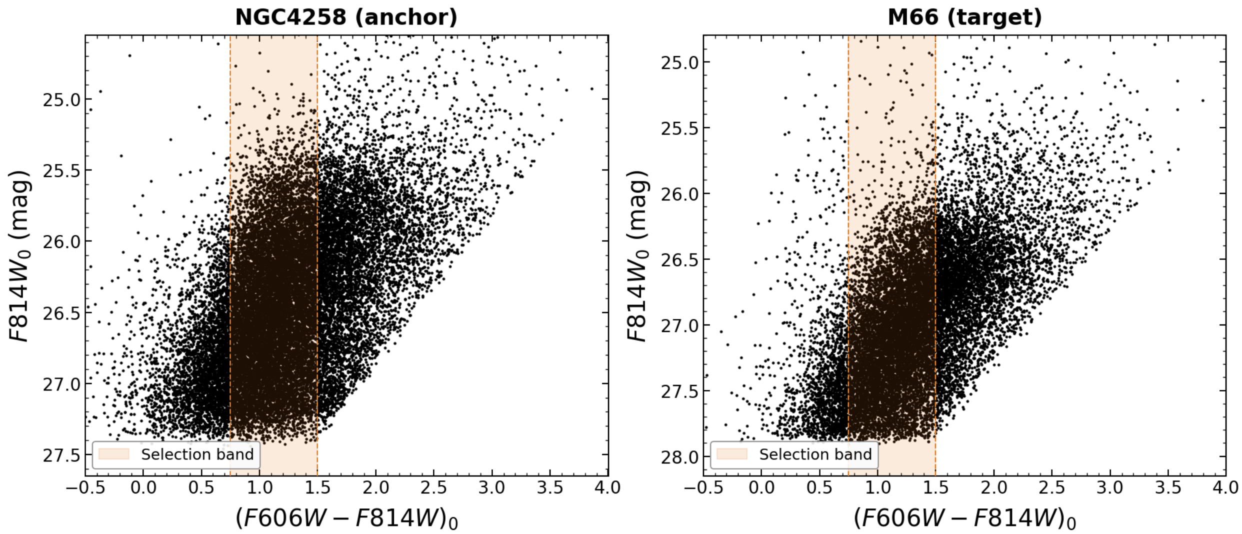
Target #12: M66 (Host of SN 1989B)

Date Processed: 2026-07-21 **Photometry Source:** EDD

1. Initial Assessment

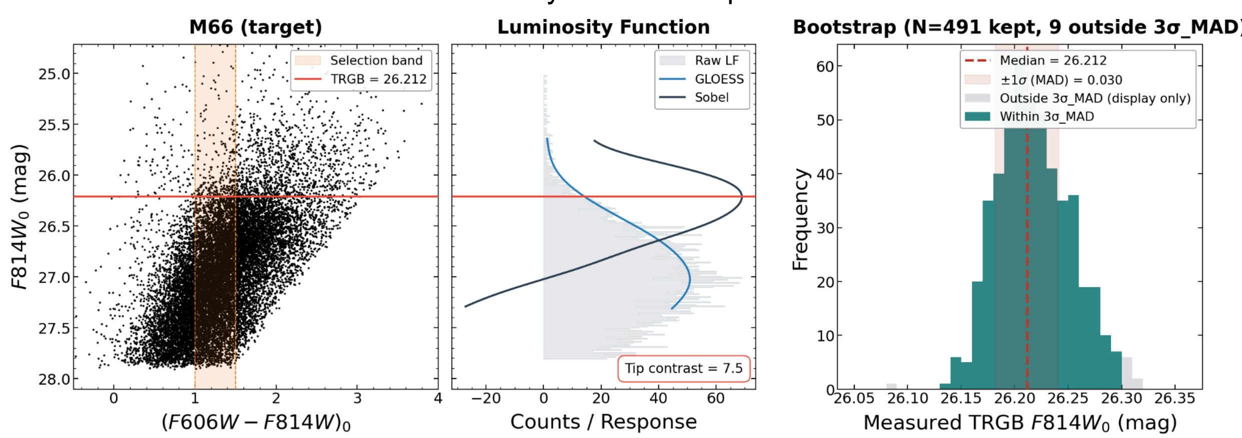
- **Data Quality:** TBD

- **Extinction:** $E(B-V) = [0.029]$



2. Parameter Choices & Justification

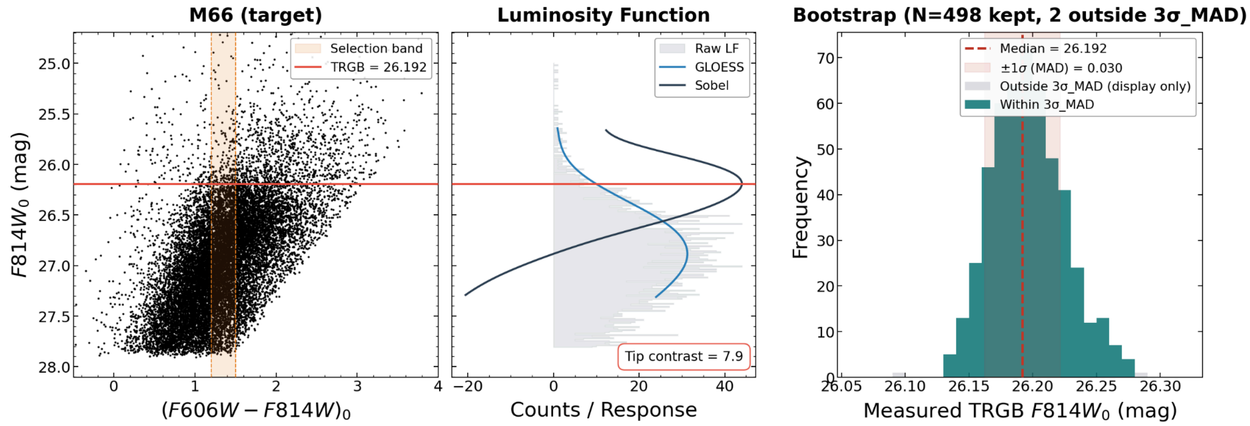
- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.2]`, `[1.5]`
  - Tightened to lower instability as much as possible
- **Spatial Clipping (`scale_factor`):** `[1]`
  - Increased to lower instability as much as possible



3. Final Diagnostics

- **Tip Contrast:** `[7.90]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.030]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** `[2631]` stars *(Target $\geq$ 400)*

- **Stability:** Slightly unstable with a shift of 0.040 over τ from 0.05 to 0.20 and a shift of 0.120 over color variations of +- 0.20 mag. *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

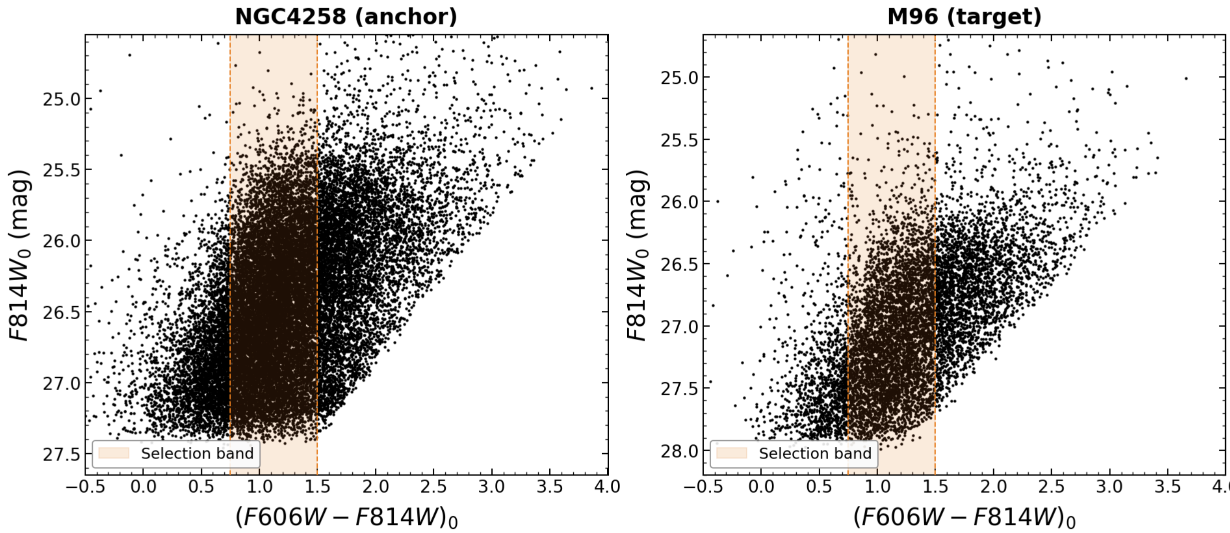
- **m_{TRGB} (Apparent):** `[26.192] ± [0.030]` mag
- **Final Distance (d):** `[10.39] ± [0.27]` Mpc

Target #13: M96 (Host of SN 1998bu)

Date Processed: 2026-07-21 **Photometry Source:** EDD

1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.022]$

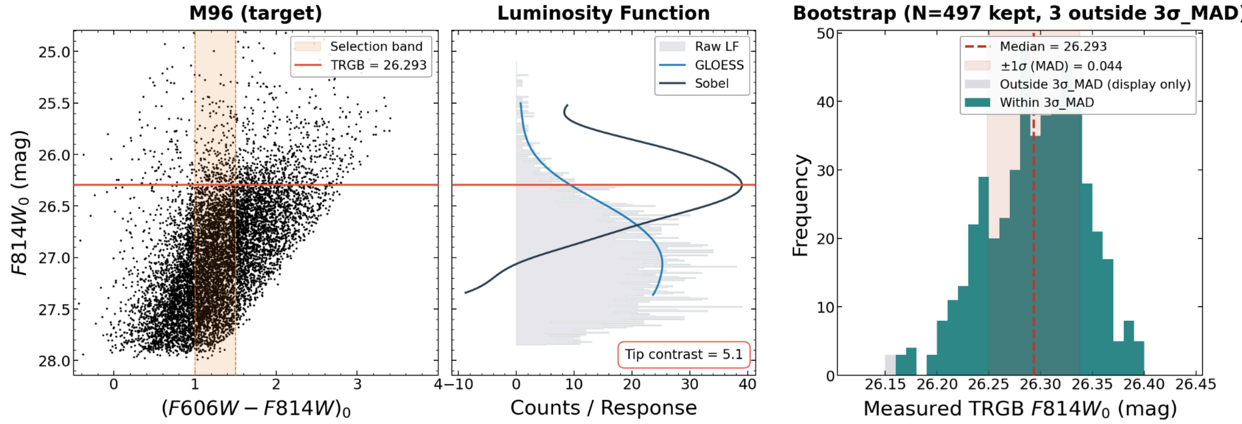


2. Parameter Choices & Justification

- **Smoothing (τ):** [0.18]
    - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** [1.0], [1.5]
    - No need to change from inherited anchor values
- **Spatial Clipping (`scale_factor`):** [0.8]
    - No need to change from inherited anchor values

3. Final Diagnostics

- **Tip Contrast:** [5.11] *(Target > 3)*
- **Bootstrap σ_{MAD}:** [0.044] mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and not too narrow
- **N_{faint}:** [2134] stars *(Target $\geq$ 400)*
- **Stability:** Somewhat unstable with a shift of 0.170 over τ from 0.05 to 0.20 and a shift of 0.130 over color variations of +- 0.20 mag. Tried to lower this by changing color_lo, color_hi, scale_factor, and adding in MAG_BRIGHT_TARGET but nothing made a substantial impact *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

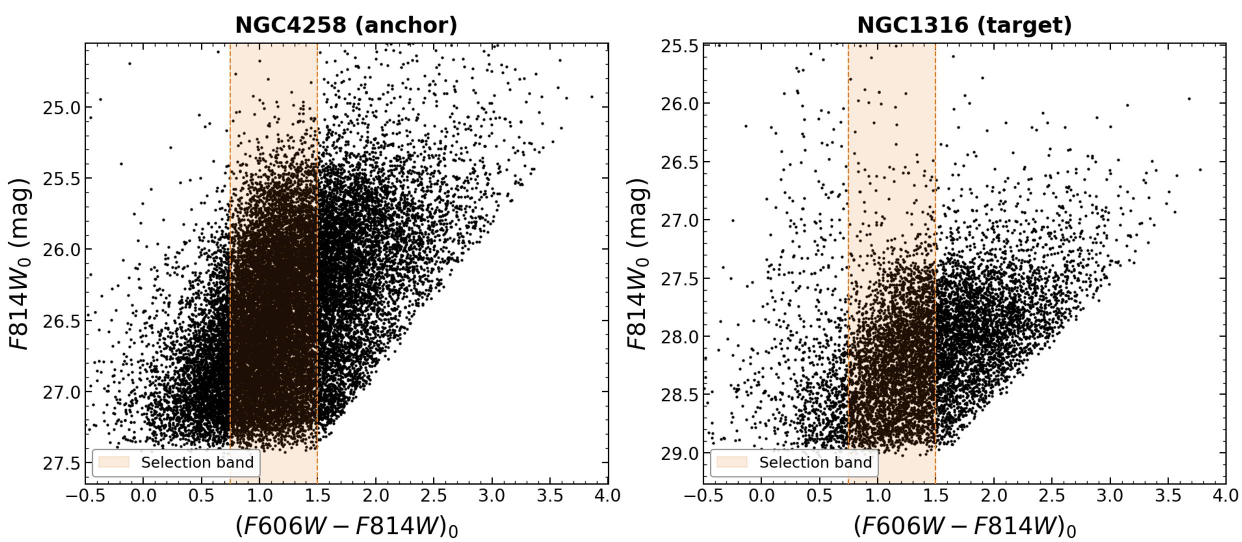
- **m_{TRGB} (Apparent):** `[26.293] ± [0.044]` mag
- **Final Distance (d):** `[10.89] ± [0.33]` Mpc

Target #14: NGC1316 (Host of SN 1980N, SN 1981D, SN 2006dd, SN 2006mr)

Date Processed: 2026-07-21 **Photometry Source:** EDD

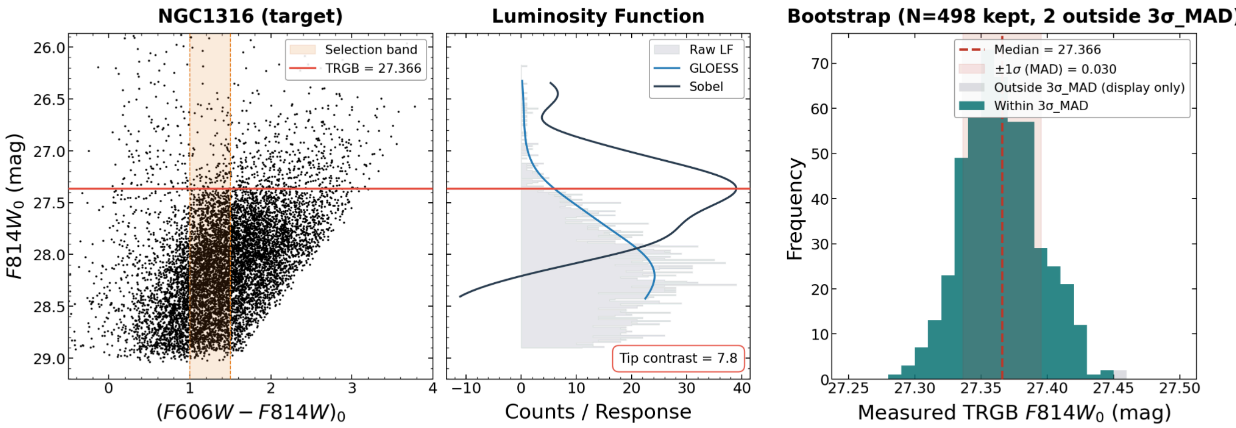
1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.018]$



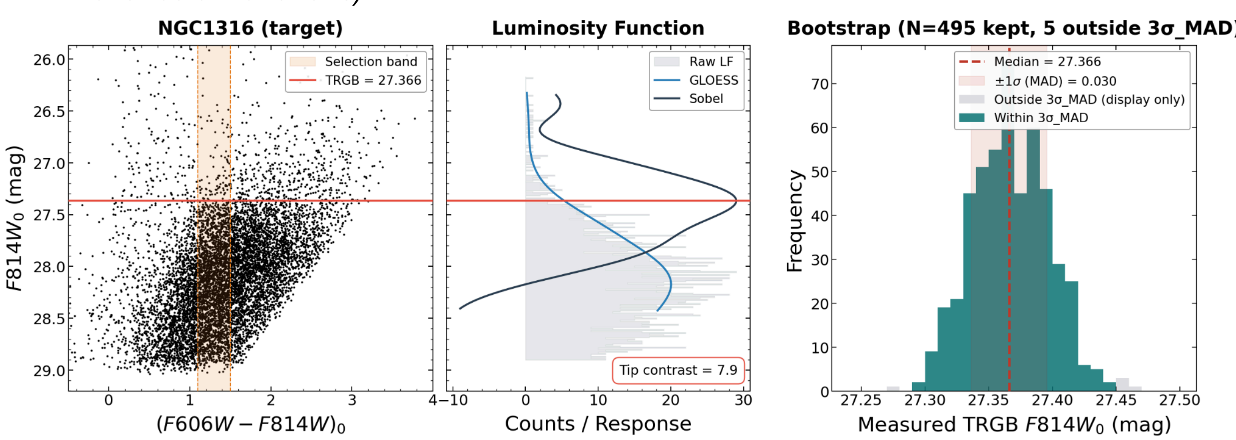
2. Parameter Choices & Justification

- **Smoothing (τ):** [0.18]
  - No need to change from inherited anchor values
- **Color Selection (color_lo, color_hi):** [1.1], [1.5]
  - Tightened to lower instability
- **Spatial Clipping (scale_factor):** [0.8]
  - No need to change from inherited anchor values



3. Final Diagnostics

- **Tip Contrast:** [7.87] *(Target > 3)*
- **Bootstrap σ_{MAD}:** [0.030] mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** [1581] stars *(Target $\geq$ 400)*
- **Stability:** Very slightly unstable with a shift of 0.030 over τ from 0.05 to 0.20 and a shift of 0.060 over color variations of +- 0.20 mag. *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

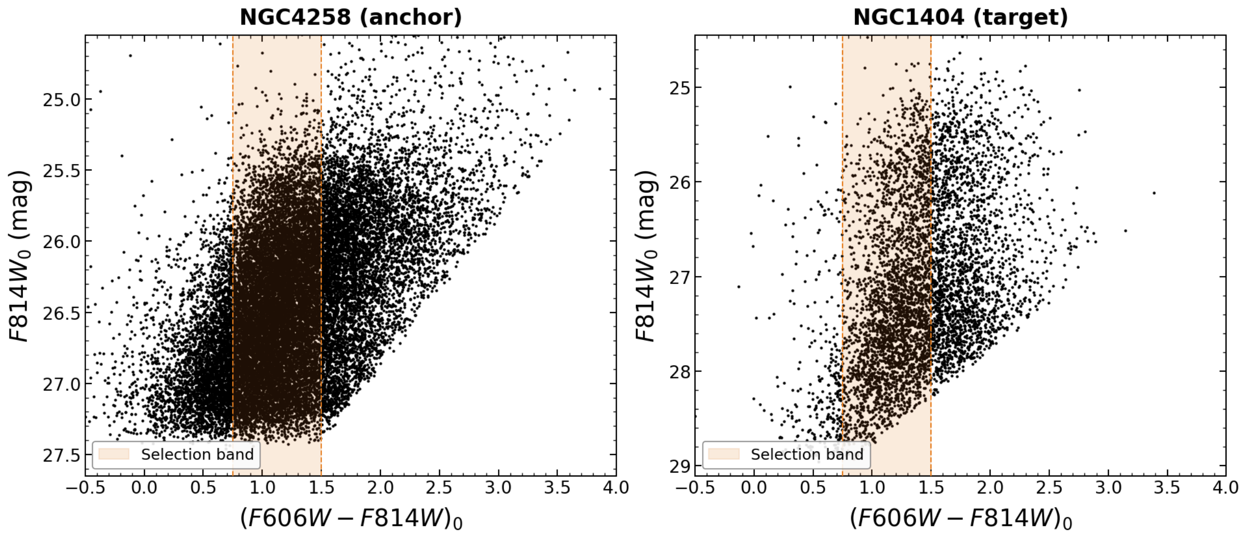
- **m_{TRGB} (Apparent):** `[27.366] ± [0.030]` mag
- **Final Distance (d):** `[17.85] ± [0.46]` Mpc

Target #15: NGC1404 (Host of SN 2007on, SN 2011iv)

Date Processed: 2026-07-21 **Photometry Source:** EDD

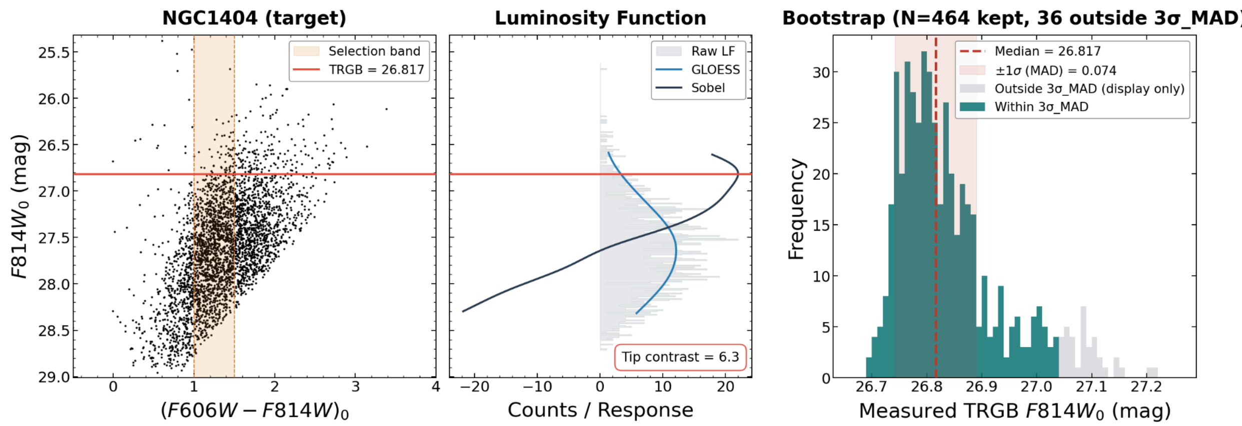
1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.010]$



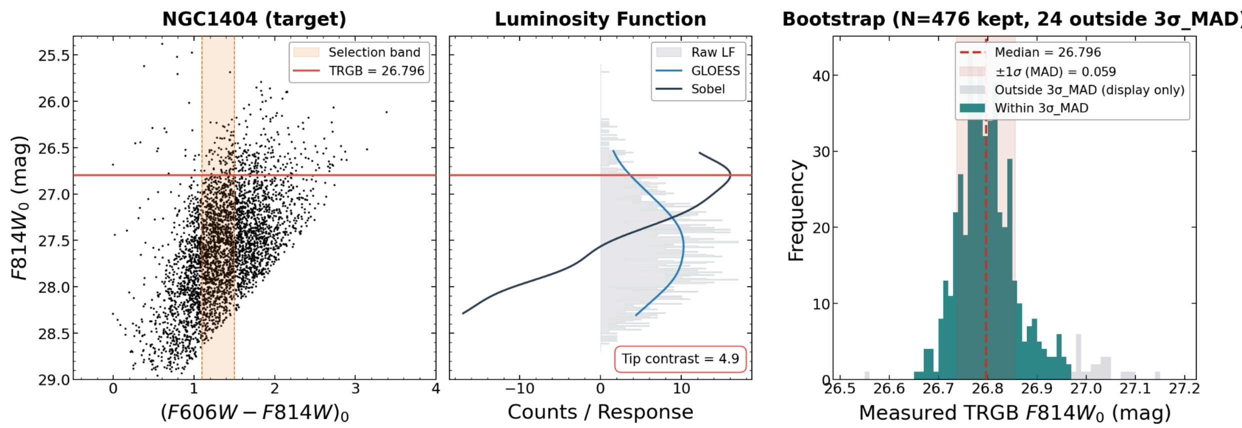
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.1]`, `[1.5]`
  - Tightened to lower instability
- **Spatial Clipping (`scale_factor`):** `[0.7]`
  - Decreased to get rid of slight bimodality on bootstrap graph



3. Final Diagnostics

- **Tip Contrast:** [4.94] *(Target > 3)*
- **Bootstrap σ_{MAD}:** [0.059] mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** [850] stars *(Target $\geq$ 400)*
- **Stability:** Slightly unstable with a shift of 0.010 over τ from 0.05 to 0.20 and a shift of 0.100 over color variations of +- 0.20 mag. *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

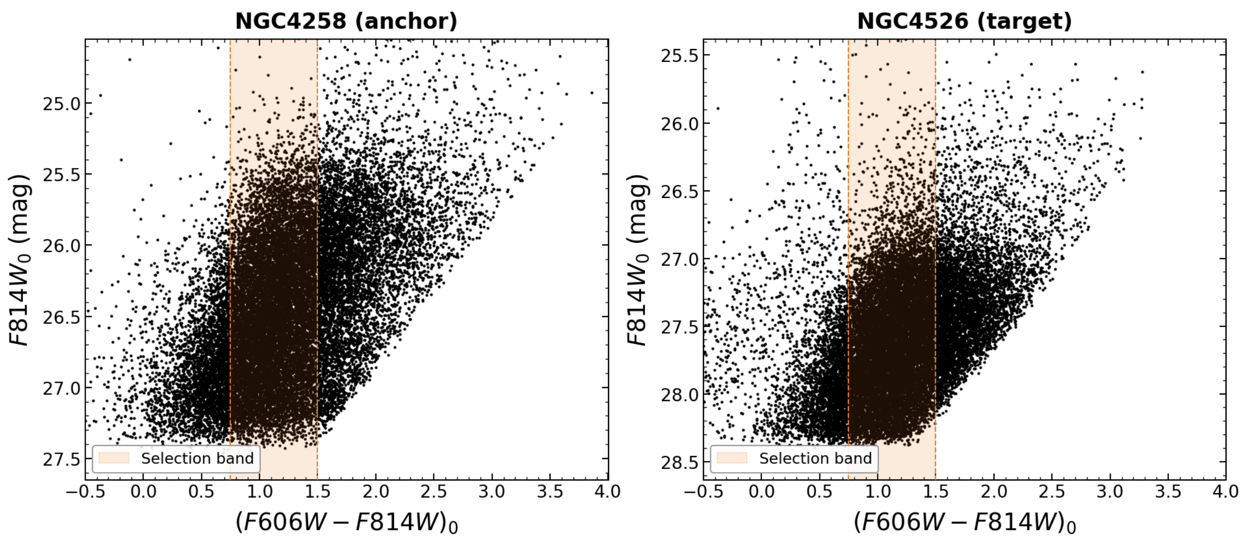
- **m_{TRGB} (Apparent):** [26.796] ± [0.059] mag
- **Final Distance (d):** [13.73] ± [0.48] Mpc

Target #16: NGC4526 (Host of SN 1994D)

Date Processed: 2026-07-21 **Photometry Source:** EDD

1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.019]$

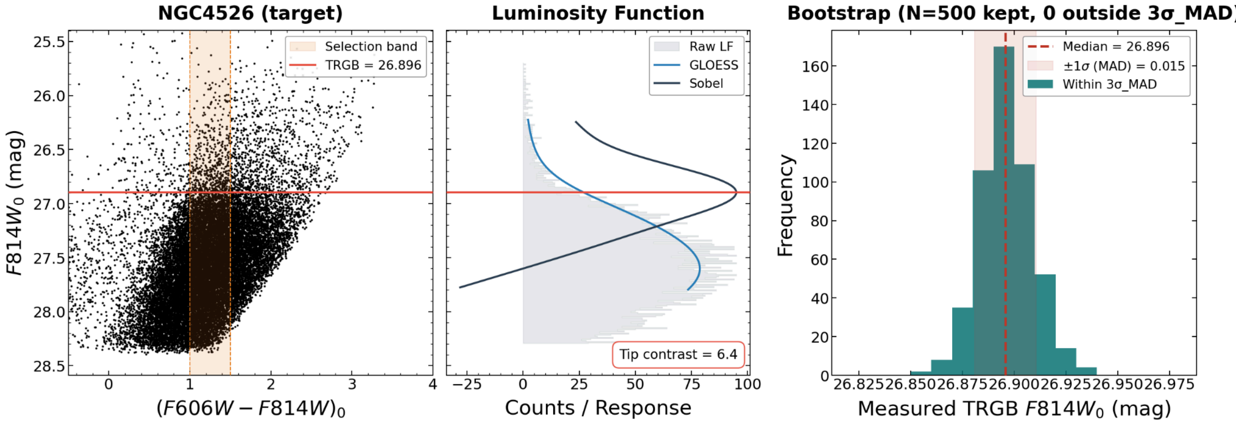


2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.0]`, `[1.5]`
  - No need to change from inherited anchor values
- **Spatial Clipping (`scale_factor`):** `[0.8]`
  - No need to change from inherited anchor values

3. Final Diagnostics

- **Tip Contrast:** `[6.37]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.015]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** `[6707]` stars *(Target $\geq$ 400)*
- **Stability:** Very slightly unstable with a shift of 0.040 over τ from 0.05 to 0.20 and a shift of 0.050 over color variations of +- 0.20 mag. *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

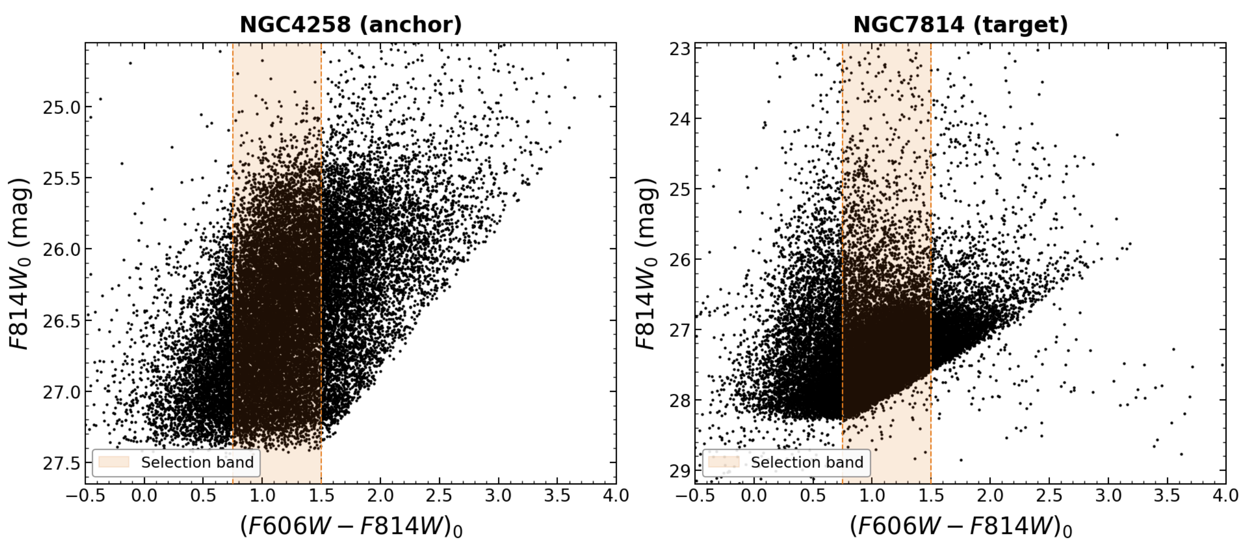
- **m_{TRGB} (Apparent):** `[26.896] ± [0.015]` mag
- **Final Distance (d):** `[14.37] ± [0.33]` Mpc

Target #17: NGC7814 (Host of SN 2021rhu)

Date Processed: 2026-07-21 **Photometry Source:** EDD

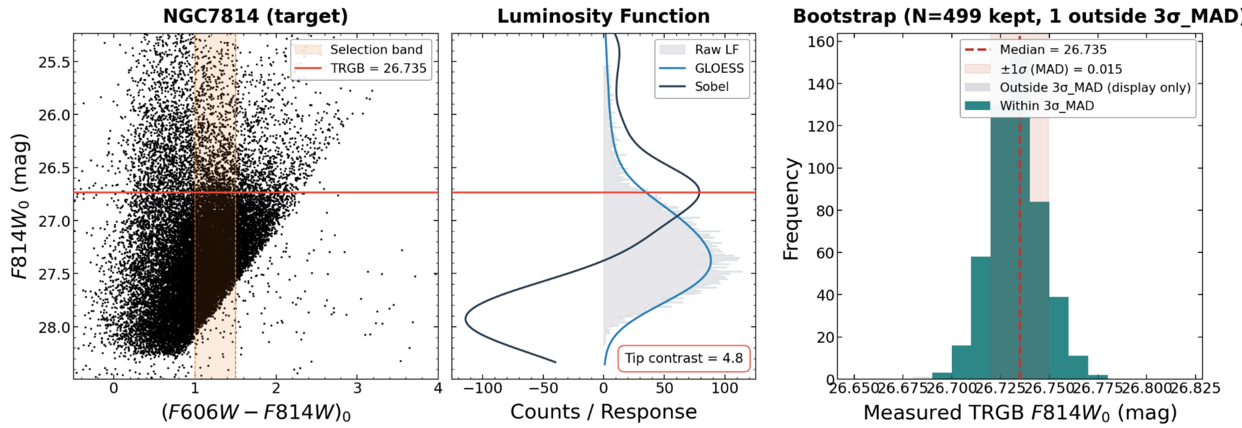
1. Initial Assessment

- **Data Quality:** TBD
- **Extinction:** $E(B-V) = [0.039]$



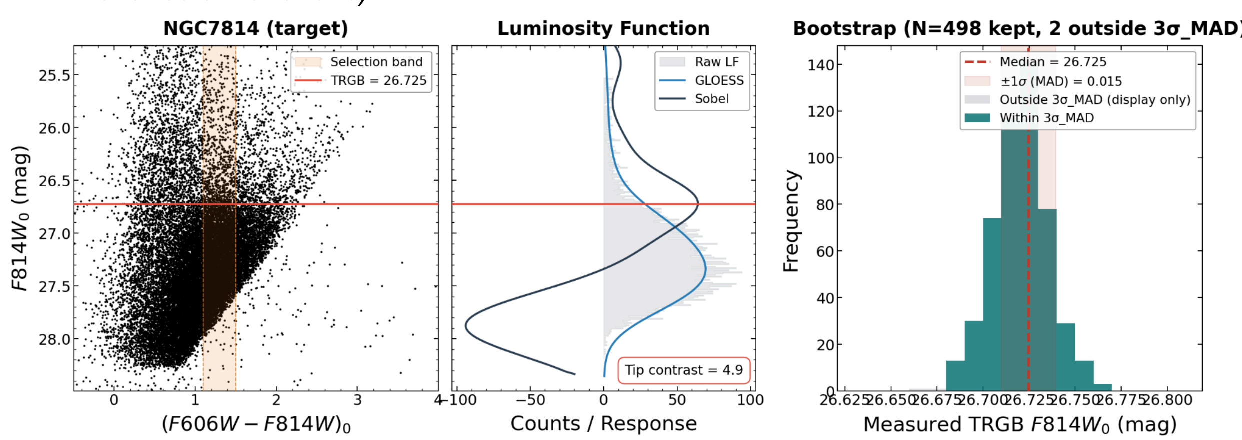
2. Parameter Choices & Justification

- **Smoothing (τ):** `[0.18]`
  - No need to change from inherited anchor values
- **Color Selection (`color_lo`, `color_hi`):** `[1.1]`, `[1.5]`
  - Tightened to lower instability
- **Spatial Clipping (`scale_factor`):** `[0.7]`
  - Decreased to lower instability



3. Final Diagnostics

- **Tip Contrast:** `[4.91]` *(Target > 3)*
- **Bootstrap σ_{MAD}:** `[0.015]` mag *(Target < 0.10)*
- **Bootstrap Shape:** Unimodal and tight
- **N_{faint}:** `[5993]` stars *(Target $\geq$ 400)*
- **Stability:** Very slightly unstable with a shift of 0.030 over τ from 0.05 to 0.20 and a shift of 0.090 over color variations of +- 0.20 mag. *(Target < 0.03 over tau range and over color variations)*



4. Distance Result

- **m_{TRGB} (Apparent):** `[26.725] ± [0.015]` mag

- **Final Distance (d):** `[13.28] ± [0.31]` Mpc

Overall H_0 from demo_mu_to_H0.ipynb

H_0 (unmodified): `[89.227] ± [1.5575]` km/s/Mpc

This is abnormally high. I believe that this is because, for target galaxies with true μ's of > 32 mag (e.g. NGC 1309, NGC 3021, NGC 3370), the Sobel filter identified the tip at m_TRGB $\approx$ 27.5 mag instead of the true TRGB tip of 28.5 mag. This leads to my μ values being capped out at 31.5 mag, and this systematically compresses the distance scale to within $\pm$ 1 mag for μ, ultimately leading to a very high H0 value.